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CADILLAC AREA GROUNDWATER CONTAMINATION
FEASIBILITY STUDY
CADILLAC, MICHIGAN

INTERIM DELIVERABLE NO. 2
ALTERNATIVES ARRAY DOCUMENT

Prepared for:
MICHIGAN DEPARTMENT OF NATURAL RESOURCES

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MEMO TO: Dianne Roycraft, MDNR

FROM: Rebecca White, E.C. Jordan Co.

DATE: July 8, 1988

SUBJECT: CADILLAC FEASIBILITY STUDY

This memorandum outlines the recent decisions made during our telephone conversations over the past two weeks, pertaining to the FS approach. Issues that remain unresolved are also presented. I anticipate that MDNR will address these issues during the comment period for the Cadillac Alternatives Array Document. The issues and the decisions made include the following:

- o At the May 26 meeting between MDNR, EPA, and E.C. Jordan, it was decided that technology-based limits would be used for the surface water discharge requirements. These limits include 20 ppb for total volatile organic compounds and 6 ppb for hexavalent chromium. However, on June 20, 1988, I received a letter from William Shaw of MDNR's Surface Water Quality Division, who indicated different technology-based standards. These standards were based on the assumption that a two-stage activated carbon treatment would be used as "model treatment". According to Mr. Shaw, different limits would be set per different technology.

As I understand, technology-based standards include the Best Available Technology (BAT) that is economically achievable. For CERCLA sites, BAT-Standards are determined on a case-by-case basis using best professional judgement (BPJ). During the BPJ evaluation, proposed remedial alternatives are reviewed to ensure the use of treatment technologies that have been proven effective to treat the pollutants present at the site. Once the BPJ determination is made, MDNR will derive numerical effluent discharge limits that are available by applying the treatment technology.

Therefore, I propose that Jordan will work with MDNR to develop water quality-based limits for a discharge to the Clam River. These limits will be compared to the treatment efficiencies of the different treatment technologies in the Detailed Analysis of Alternatives. Alternative not attaining the water quality-based limits will be screened out during the detailed analysis.

Beth Ryan of E.C. Jordan, will call you to set-up a contact with someone from MDNR's Surface Water Quality Division, so that the water quality based limits may be developed.

- o At the May 26 MDNR/EPA/E.C. Jordan meeting, it was decided that the preliminary target levels for soils would be used as final target levels except for xylene. However, in a June 17, 1988 MDNR memorandum (from Brad Venman to Ev Bole), Brad recommended using a linkage model, in conjunction with the OLM model, to estimate groundwater concentrations below a waste deposit at the water table.

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Using this model could potentially change the target levels. Jordan is awaiting a decision by MDNR on the appropriateness of using this model. If the model is used and target levels change, alternatives that do not attain those levels will be screened out in the Detailed Analysis of Alternatives.

- o Additional soil sampling will be conducted at Leo Ingraham's by MDNR and at Rexair by another consultant engaged by Rexair. If chemicals are detected in soils at either of these areas, at concentrations above the specific soil target levels calculated for the same chemicals at Kysor or Four Winns, Jordan will conduct an analysis of remedial alternatives for those areas as an addendum to the FS, requiring approximately 40 additional hours.

If there are any errors or comments regarding this report, please contact Bob Koltuniak at 313-661-3100. Comments on the alternatives Array Document should also be directed to Bob Koltuniak. Thank you.



Rebecca White
Scientist

RW:hpm

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1.0 INTRODUCTION

Under contract to the Michigan Department of Natural Resources (MDNR), E.C. Jordan Co. (Jordan) is presently conducting a Remedial Investigation/Feasibility Study (RI/FS) of the contamination (primarily in groundwater) beneath the Cadillac Industrial Park in Cadillac, Michigan. Because the Northernaire Plating Company (Northernaire) and Kysor of Cadillac, Inc. (Kysor) Superfund sites are located within the study area, the study is being conducted under joint MDNR-U.S. Environmental Protection Agency (USEPA) guidance, and in accordance with the Comprehensive Environmental Response, Compensation, and Liability Act of 1980 (CERCLA), as amended by the Superfund Amendments and Reauthorization Act of 1986 (SARA), and the National Oil and Hazardous Substances Pollution Contingency Plan (NCP 1985 - Final Rule, Section 300.68). The Northernaire site and associated metals contamination were the subjects of an RI and a focused FS for contaminated site soils, also performed by Jordan for MDNR. Because the Northernaire plume was known to mingle with other source contaminants, it was decided that the Northernaire operable unit for the groundwater would be performed in conjunction with the Cadillac FS, which also includes assessments of volatile organic compound (VOC) contamination from Kysor and other identified sources in the Cadillac Industrial Park.

Investigation and remediation of site contamination for the Cadillac site has been divided into two parts, soil and groundwater. The RI presented the distribution of VOC and chromium contamination in groundwater beneath the Cadillac Industrial Park and adjacent areas of Haring Township, Michigan, and

identified some residual contaminants in soils at suspected source locations. Information sources for the RI included studies performed by Jordan, MDNR, and USEPA, and by consultants for industries within the industrial park. A baseline risk assessment (RA) was then prepared based on the findings of the RI. The FS will identify and analyze alternatives for groundwater remediation of the contaminated plumes and soil remediation of the identified source locations.

An interim deliverable, presenting the remedial action objectives for the Cadillac FS, including preliminary target levels and discharge requirements, was submitted to MDNR and USEPA in April 1988. Based on MDNR-USEPA comments on the remedial action objectives for the Cadillac site, Jordan developed final target levels for groundwater and soil remediation and revised the requirements for discharge to surface water to reflect the Michigan water quality standards. These target levels form the basis for this Alternatives Array Document.

1.1 PURPOSE AND SCOPE

The purpose of the Alternatives Array Document is to develop and identify those alternatives that will achieve the remedial action objectives. The development of alternatives involves identifying applicable technologies and screening out those that would not meet response objectives or which are technically infeasible due to specific site characteristics, waste characteristics, or other physical constraints.

Technologies that pass the screening are then combined to form a range of medium-specific alternatives. SARA emphasizes the use of treatment technologies that reduce toxicity, mobility, or volume of contaminants rather than technologies that prevent exposure. Therefore, the range includes, to the extent possible, alternatives from each of the following categories:

- o no action
- o containment
- o treatment

The treatment alternatives range from those that eliminate the need for long-term site management to those that reduce toxicity, mobility, or volume of contaminants. The treatment alternatives, once developed, are screened based on specific criteria pertaining to effectiveness, implementability, and cost. Summaries of the alternatives retained for both groundwater and soil remediation are presented in Sections 4.4 and 5.4, respectively.

Upon MDNR-USEPA concurrence with the remedial alternatives remaining after initial screening, Jordan will conduct a detailed analysis of each alternative. In this phase, alternatives will be described and analyzed against several factors including the following:

- o compliance with ARARs;
- o reduction of the toxicity, mobility, or volume of contaminants;
- o short-term effectiveness;
- o long-term effectiveness and permanence;

- o overall protection of human health and the environment;
- o implementation;
- o community and state acceptance; and
- o costs and economic analysis.

Following detailed analysis, the results will be summarized and compared considering the following factors:

- o protection of human health and the environment;
- o attainment of federal and state public health and environmental requirements identified for the site;
- o cost effectiveness; and
- o use of permanent solutions and alternative treatment technologies or resource recovery technologies, to the maximum extent practicable.

The completed FS will serve as the basis for the Record of Decision (ROD), which documents remedy selection, and for remedial design and construction.

1.2 SITE DESCRIPTION

The Cadillac Industrial Park includes nearly 40 industrial facilities. Past uncontrolled releases of contaminants to the ground by several of these facilities have contaminated groundwater in aquifers underlying the park. The

contaminants of concern are VOCs (i.e., chlorinated and aromatic hydrocarbons such as trichloroethylene [TCE] and xylene) and hexavalent chromium (derived from chromium plating wastes). The presence of contaminants within the upper aquifers beneath the site has created concern for potential impact on a lower aquifer. This aquifer supplies potable water to the city from a well field located near the center of the industrial park.

The RI identified five contaminant sources and seven contaminant plumes in groundwater in the study area. The five probable or known source locations are (1) Northernnaire; (2) Kysor; (3) Rexair (probable); (4) Four Winns; and (5) Four Star/Joynts (former).

In the following descriptions, the precise units for concentration, milligrams per liter (mg/l), have been replaced by parts per million (ppm), which are conceptually more easily appreciated. At the low concentrations detected, the concentrations are numerically equal for both sets of units.

The seven plumes are categorized under four major plumes and include the following:

Northernnaire Plumes: The following two plumes make up the Northernnaire Plumes:

- (1) A hexavalent chromium plume in the intermediate aquifer situated just above the lower clay aquitard. Northernnaire was the source.

- (2) A small hexavalent-chromium plume of relatively low concentration in the shallow aquifer, interpreted as a small slug released from Northernnaire. The slug, moving rapidly through the upper aquifer and being diluted as it migrates, should soon merge with the intermediate aquifer.

Shallow VOC Plume: a large, concentrated VOC plume (up to 115 ppm) in the shallow aquifer caused primarily by past solvent dumping behind Kysor. This plume has merged with other VOC releases from Four Star/Joynts (former) to the northeast and Four Winns to the northwest.

Intermediate VOC Plume: a large VOC plume (up to 12 ppm) in the intermediate aquifer resulting from Kysor-related VOCs migrating from the shallow aquifer through the upper aquitard.

Rexair Plume: a relatively highly concentrated VOC plume (up to 153 ppm) at depths to 140 feet, located in the intermediate aquifer between Rexair and the city well field.

Other Plumes. The RI also identified two small VOC plumes in the North Park Subdivision. These plumes are located horizontally downgradient of the city well field and appear to pose a less immediate threat (if any) to the city water supply. It is believed that all potential receptors (i.e., domestic wells) downgradient of these plumes were replaced by municipal water supplies.

Therefore, these plumes were not assessed as a potential human health risk in the baseline RA. However, remediation of these two plumes is included in the FS as a possible, additional component of each groundwater alternative.

Soils. Only one area of soils, the former solvent disposal area behind Kysor, was positively identified in the RI as having significant residual VOC contamination. This area appears to be a continuing source of groundwater contamination in the shallow and intermediate aquifers southwest of the city well field. In addition, soils at Four Winns are assumed to be a potential continuing source of groundwater contamination. However, because Four Winns, Inc. elected to participate as a potentially responsible party in the study with its own consultant, Jordan was not able to assess soil contamination at Four Winns as a source of groundwater contamination. Based on 1985 data from MDNR, Jordan has concluded that concentration levels in Four Winns soils could require remediation. Other sources of soils contamination could be present at the Ingraham property and at Rexair. Studies further defining both plumes and sources at Rexair are being conducted by a consultant engaged by Rexair. Other explorations at Ingraham will be performed by MDNR under another contract to MDNR. If chemicals are detected in soils at the Ingraham property or at Rexair, at concentrations above the specific soil target levels calculated for the same chemicals at Kysor or Four Winns, Jordan will include an analysis of remedial alternatives for those areas as an addendum to the FS.

2.0 SUMMARY OF REMEDIAL ACTION OBJECTIVES

Remedial action objectives and site-specific considerations must be identified to select the appropriate remedial technologies and formulate remedial alternatives. Remedial action objectives are statements of the proposed level of cleanup associated with the specific chemical contamination problems in the various site media. Applicable or Relevant and Appropriate Requirements (ARARs) that establish clean-up standards are also used to determine remedial action objectives. This section summarizes the remedial action objectives identified in the first FS interim deliverable submitted to MDNR and USEPA in April 1988. The completed draft FS will include the complete, detailed discussion of remedial action objectives.

Discharge requirements for potential discharges of treated groundwater to surface water, publicly owned treatment works (POTWs), or recharge to groundwater are also presented in this section; the requirements are used to determine the appropriate level of treatment before discharge.

2.1 GENERAL RESPONSE OBJECTIVES

In the RA, the only exposure pathway identified as presenting a potential threat to public health is the domestic use of contaminated groundwater (i.e., drinking, showering, bathing, and cooking). Because each contaminated groundwater plume identified and evaluated in the RA is considered an existing

or potential source of drinking water, the following separate remedial action objectives are proposed for each plume:

Shallow VOC Plume. Reduce human health risks posed by potential use of the shallow groundwater plume as a domestic water supply (including drinking). The shallow plume is contaminated with 1,1,1-trichloroethane (1,1,1-TCA); 1,1-dichloroethylene (1,1-DCE); TCE; 1,2-dichloroethane (1,2-DCA); and tetrachloroethylene (PCE).

Intermediate VOC Plume. Reduce human health risks posed by the potential use of the intermediate groundwater plume (which is contaminated with trans-1,2-dichloroethylene [t-1,2-DCE]; TCE; 1,1-DCE, and PCE) as a domestic drinking water supply.

Northernaire Plumes. Reduce human health risks posed by the potential use of the hexavalent chromium-contaminated groundwater in the Northernaire Plumes as a domestic drinking water supply.

Rexair Plume. Reduce potential human health risks posed by using the Rexair groundwater plume, primarily contaminated with TCE and methylene chloride (MC), as a domestic drinking water supply.

As discussed in the RA, the residual soil contamination at Kysor and Four Winns does not pose a direct contact threat because the contaminated soil is subsurface. However, the RA did assess residual soil contamination at Kysor as a continuing source of groundwater contamination, based on the detected

concentrations of the contaminants and their ability to leach to groundwater. Adequate data were not available to assess existing soil contamination at Four Winns as a source of groundwater contamination. However, based on 1985 data, it is assumed that concentration levels at Four Winns are high enough to warrant remediation. Based on this assessment, the following remedial action objectives are proposed:

Kysor. Reduce potential human health risks posed by the continued leaching of TCE, toluene, and 1,1,1-TCA from subsurface soils to a potential drinking water source. Reduce potential human health risks posed by the potential for xylene to leach from subsurface soils to a potential drinking water source.

Four Winns. Reduce potential human health risks posed by the potential for continued leaching of 1,1,1-TCA, toluene, TCE, and MC from subsurface soils to a potential drinking water source.

2.2 SUMMARY OF CHEMICAL-SPECIFIC ARARS

As discussed in the Cadillac RI and the first FS interim deliverable, ARARs are federal and state public health and environmental requirements and guidelines that apply to hazardous waste site cleanup. Section 121 of CERCLA, amended by SARA, requires that hazardous waste remedial actions comply with federal ARARs and state ARARs that are more stringent, legally enforceable, and consistently enforced statewide. ARARs are used to determine the appropriate extent of site

cleanup, to scope and formulate remedial action alternatives, and to govern the implementation and operation of the selected action.

As discussed in the RI, ARARs may be characterized as chemical-, location-, or action-specific. Chemical-specific requirements are used to determine the remedial action objectives because they set health- or risk-based concentration limits or discharge limitations in various environmental media for specific hazardous substances, pollutants, and contaminants. Chemical-specific ARARs for the Cadillac site were identified and discussed in the RI. A list of all chemical-specific ARARs and policy and guidance to be considered (TBCs) for the Cadillac site, and a synopsis of the requirements, are contained in Appendix A. Actual contaminant concentration levels for the groundwater required by ARARs, which are used to set groundwater remedial action objectives, are listed in Appendix B, and are compared to the maximum concentrations of the groundwater chemicals identified as contributing the most to total risk in each plume. Because contaminants in the soil are presently leaching or have a potential to leach into groundwater, concentrations of chemicals which are presently not a problem in groundwater, but which are found at high concentrations in soil, are also compared to groundwater ARARs.

2.3 SUMMARY OF TARGET LEVELS FOR GROUNDWATER REMEDIATION

Groundwater target clean-up levels (TCLs) are the chemical concentrations in each plume that would be protective of human health if long-term exposures were to occur. The baseline RA for the Cadillac site (Section 8.0 of the RI)

provides a basis for determining which chemicals of concern should be targeted for cleanup in each plume. It is not necessary to calculate target levels for every chemical found in the plumes. Cleanup should focus on those chemicals that have been shown in the RA to pose "significant" risks to public health (i.e., carcinogenic risks within or above the target range of 10^{-4} to 10^{-7} , or noncarcinogenic risks greater than one).

For the Cadillac area groundwater contamination, the chemicals in each plume that present "significant" risks are as follows:

- o Shallow VOC Plume:

- 1,1,1-TCA

- 1,1-DCE

- TCE

- Toluene

- 1,2-DCA

- PCE

- o Intermediate VOC Plume:

- c-1,2-DCE

- TCE

- 1,1-DCE

- PCE

- o Rexaire Plume:

- TCE

- MC

- o Northernnaire Plume:

Hexavalent Chromium (Cr[+6])

Preliminary target levels were calculated for carcinogens on the basis of public health risk calculations. Preliminary target levels for noncarcinogens were defined on the basis of health-based ARARs or, if ARARs do not exist, public health risk calculations. Several options were developed for each groundwater plume that incorporate risk levels, ARARs, or a combination of the two.

The preliminary groundwater TCLs were then revised as part of the FS process in response to MDNR policy. The final TCLs achieve a specified total risk range and reflect public health, environmental, engineering, and policy considerations. In general, MDNR policy requires that TCLs for carcinogens are set as close to the 10^{-6} risk level as technically possible. However, in many cases, a 10^{-6} risk level is below practical detection limits and/or too low to achieve by best available technology (BAT). Therefore, the revised TCLs, known as the final groundwater TCLs for the Cadillac area groundwater contamination, were set as close to the 10^{-6} risk level as technically possible, using a combination of ARARs, risk-based levels, and/or the Contracted Required Detection Levels (CRDLs). USEPA Region V was consulted and concurred with this decision.

Final target levels for groundwater remediation at the Cadillac site are listed in Table 2-1.

TABLE 2-1

CADILLAC AREA GROUNDWATER CONTAMINATION
FINAL GROUNDWATER TARGET LEVELS

CADILLAC AREA FEASIBILITY STUDY

PLUME	CHEMICAL	TARGET LEVEL (mg/l)	REFERENCE
Shallow	1,1,1-Trichloroethane	0.2	MCL
	Toluene	2	MCLG
	1,1-Dichloroethylene	0.005	CRDL
	Trichloroethylene	0.005	MCL
	1,2-Dichloroethane	0.005	MCL
	Tetrachloroethylene	0.001	CALCULATED
Intermediate	Trans-1,2-dichloroethylene	0.07	MCLG
	1,1-Dichloroethylene	0.005	CRDL
	Trichloroethylene	0.005	MCL
	Tetrachloroethylene	0.001	CALCULATED
Rexair	Methylene chloride	0.005	CALCULATED
	Trichloroethylene	0.005	MCL
Northernnaire	Chromium hexavalent	0.05	MCL

NOTES:

CRDL = Contract Required Detection Level.

MCL = Maximum Contaminant Level.

MCLG = Maximum Contaminant Level Goal.

2.4 SUMMARY OF TARGET LEVELS FOR SOIL REMEDIATION

The general response objective for remediation of soils at the Cadillac Industrial Park is to reduce residual contamination to levels that will no longer significantly contribute to the groundwater contamination.

The first step in the development of soil target levels is to identify the chemicals in each area that are, or have the potential to, significantly impact groundwater. This was accomplished for Kysor soils by identifying the chemicals of concern that (1) are present in the soils at high concentration, and (2) are posing "significant" risks (or are at levels above ARARs) in the shallow plume. Using these criteria, TCE, toluene, and 1,1-TCA are included in the TCL evaluation. Xylene was also included because it was found at the highest concentrations in the Kysor soils. Available data are inadequate to assess Four Winns soils. However, soil TCLs are developed for those chemicals that may be in the soils at significant concentrations based on 1985 and limited 1987 data. It is assumed that because the 1985 data were taken from sampling events conducted closer to the time of the discharge to the ground, the chemicals detected in the soils would not have increased in concentration over time, but would likely have decreased, if they have changed at all. These chemicals include 1,1,1-TCA, toluene, TCE, and MC in the Four Winns soils.

There are currently no health-based ARARs for these chemicals in soils. Therefore, all preliminary soil target levels were calculated on the basis of risk. This was accomplished by using the final groundwater target levels identified for these chemicals in Section 2.3 and back calculating to a "safe"

concentration in the soil using the defining equation for the USEPA Organic Leaching Model (OLM).

The following subsections summarize the identification of target levels for soils in Cadillac Industrial Park.

Kysor. The parameters necessary to calculate target levels for chemicals in the Kysor soils, and the TCLs themselves, are summarized in Table 2-2. Xylene did not have a preliminary groundwater target level, as it was not identified as posing a significant risk in the baseline RA. Therefore, a health-based groundwater TCL was identified for xylene based on its proposed Maximum Contaminant Level Goal (MCLG) of 0.44 mg/l. TCLs for Kysor soils range from 0.07 ppm for TCE to 724 ppm for toluene.

Four Winns. Table 2-3 summarizes the parameters and preliminary TCLs for chemicals in the Four Winns soils. TCLs for Four Winns soils range from 0.02 ppm for MC to 724 ppm for toluene.

2.5 TREATED GROUNDWATER DISCHARGE REQUIREMENTS

Potential options for the disposal of treated groundwater include discharge to groundwater, surface water, or a POTW. Discharge requirements for these options are identified in Table 2-4, using the chemicals identified in the RA

CADILLAC AREA GROUNDWATER CONTAMINATION
 TARGET LEVELS
 KYSOR SOILS
 CADILLAC AREA FEASIBILITY STUDY

CHEMICAL	GROUNDWATER TARGET LEVEL (mg/l)	SOLUBILITY (S, mg/l)	SOIL TARGET LEVEL (MG/KG)
Trichloroethylene	0.005	1100	0.07
Toluene	2	535	724
Xylene	0.44	180	141
1,1,1-Trichloroethane	0.2	4400	7.6

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TABLE 2-3
 CADILLAC AREA GROUNDWATER CONTAMINATION
 TARGET LEVELS
 FOUR WINNS SOILS
 CADILLAC AREA FEASIBILITY STUDY

CHEMICAL	GROUNDWATER TARGET LEVEL (mg/l)	SOLUBILITY (S, mg/l)	SOIL TARGET LEVEL (MG/KG)
Trichloroethylene	0.005	1100	0.07
1,1,1-Trichloroethane	0.2	4400	7.6
Toluene	2	535	724
Methylene chloride	0.005	16700	0.02

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TABLE 2-4

TREATED GROUNDWATER DISCHARGE REQUIREMENTS
CADILLAC AREA FEASIBILITY STUDY

COMPOUND	MAXIMUM CONCENTRATION DETECTED (mg/l)	DISCHARGE TO GROUNDWATER (mg/l)	REFERENCE	DISCHARGE TO SURFACE WATER** (mg/l)	REFERENCE	DISCHARGE TO CADILLAC POTW ¹ (mg/l)
1,1,1-Trichloroethane	32.0	0.2*	MCL	≤ 0.005	CRDL	.1
Trans-1,2-dichloroethylene	4.6	0.07*	Proposed MCLG	≤ 0.005	CRDL	.1
1,1-Dichloroethylene	1.0	0.005*	CRDL	≤ 0.005	CRDL	.1
1,2-Dichloroethane	2.0	0.005*	MCL	≤ .005	CRDL	.1
Methylene chloride	0.69	0.005	Calculated	≤ 0.005	CRDL	.1
Tetrachloroethylene	0.72	0.001	Calculated	≤ 0.005	CRDL	.1
Trichloroethylene	153.0	0.005*	MCL	≤ 0.005	CRDL	.1
Toluene	3.2	2.0	MCLG	≤ 0.005	CRDL	
Chromium (hexavalent)	0.773	0.05*	MCL	≤ 0.010	CRDL	.355 (total chromium)

NOTES:

* A non-detectable level may be required under MGWQ, Part 22.

** May vary when discharge requirements are finalized.

¹ Reference - Cadillac POTW Pretreatment Program Requirements.

MCL = Maximum Contaminant Level.

CRDL = Contract Required Detection Level.

as contributing the most to the total risk in each plume. Although other contaminants were detected in groundwater in past sampling episodes, remediation of the specified compounds identified from the RA to discharge limits should remove other compounds that may be present to acceptable levels.

While the Safe Drinking Water Act (SDWA) Maximum Contaminant Levels (MCLs) and MCLGs and the Clean Water Act (CWA) Ambient Water Quality Criteria (AWQC) all establish clean-up levels for public drinking water systems, the Michigan Water Resources Commission has established groundwater quality rules for discharges to groundwater, under Michigan Groundwater Quality Rules (MGWQ), Part 22. Generally, these discharge rules are more stringent than MCLs and AWQC, and maintain that the quality of groundwater in all usable aquifers shall not be degraded from local background groundwater quality as the result of a discharge. Compounds at concentrations that exceed MCLs shall not be discharged into groundwater in usable aquifers even in those cases where the local background groundwater levels for these compounds exceed the MCLs. Generally, Michigan's nondegradation policy for groundwater discharges prohibits discharge of organic contaminants at levels above detection. Variances may be granted by the commission for exceptional circumstances where it is determined that (1) strict conformity with the rule is not economically or technically feasible; (2) a prudent alternative does not exist; and (3) such variances are consistent with the promotion of the public health, safety, and welfare. Therefore, requirements for discharges to groundwater, presented in Table 2-4, are based on MCLs, final target levels, or nondetectable levels.

For surface water discharges, the Michigan Water Quality Standards are applicable. Under these standards, toxic substances shall not be present in the waters of the state at levels that are or may become injurious to the public health, safety, or welfare; plant and animal life; or the designated uses of those waters. Allowable levels of toxic substances in discharges to surface water shall be determined by the MDNR using technology-based standards developed by MDNR pursuant to R323.2137 and R323.2140 and imposed under Section 402(a)(1) of the CWA. Water quality-based effluent limits shall also be used, but only when they are more restrictive than the technology-based limits.

Technology-based standards include, for toxic and nonconventional pollutants, the BAT that is economically achievable. Technology-based requirements are determined through effluent limitations for specific categories of industries. For CERCLA sites, BAT-based standards are determined on a case-by-case basis using best professional judgement (BPJ). During a BPJ evaluation, proposed remedial alternatives will be reviewed to ensure the use of treatment technologies that have been proven effective to treat the pollutants present at the site. Once this BPJ determination is accomplished, the National Pollutant Discharge Elimination System (NPDES) permitting authority will derive numerical effluent discharge limits that should be achievable by applying the treatment technology.

Under the direction of MDNR, Jordan will develop water quality-based limits for a discharge to the Clam River. These limits will be compared to the treatment efficiencies of the different treatment technologies in the Detailed Analysis

of Alternatives. Alternatives not attaining the water quality-based limits will be screened out during the detailed analysis.

In the absence of water quality-based limits, the technology-based limit approach will be used, and technologies applicable to the pollutants at the site will be identified. The compound most difficult to remediate by a specific technology will dictate the system design in order to treat to the lowest technically feasible level. Because this approach allows for the lowest most practicable reduction of contaminant levels, it is assumed these levels will be as stringent if not more stringent than water quality-based effluent limits. The preliminary surface water discharge requirements, presented in Table 2-4, represent levels at or below detection levels.

If groundwater is discharged to a POTW, it must meet pretreatment standards set by the POTW. The City of Cadillac POTW does have an approved pretreatment program with specific limits for the contaminants listed in Table 2-4. If the Cadillac POTW is utilized for disposal, the groundwater will be pretreated to meet appropriate standards.

3.0 IDENTIFICATION AND SCREENING OF TECHNOLOGIES

Based on the response objectives and present understanding of the groundwater and soil contamination, treatment technologies suitable to conditions at the Cadillac Industrial Park site have been identified. In the identification process, the demonstrated performance of each technology is considered. Those technologies that are only conceptual in development are not identified. However, to the extent possible, innovative technologies are identified (i.e., technologies that have not been proven in hazardous waste application or have been proven only on a pilot scale).

After technologies are identified, they are screened based on engineering feasibility and implementability. The screening consists of evaluating technologies to ensure that they are suitable for the location and conditions at the Cadillac site, and applicable to the contaminants and environmental media. Therefore, the assessment considers both waste and site characteristics.

Waste characteristics include physical properties of the waste (e.g., volatility, solubility, chemical concentrations, and other properties of specific chemical constituents that affect the performance of a technology). Site characteristic information gathered during the RI (e.g., site geology and hydrogeology) are reviewed to identify conditions that may limit or favor the use of certain remedial technologies. Site characteristics also include the availability of space and resources necessary to implement the technology.

Those technologies whose use is clearly precluded by waste or site characteristics were eliminated from further consideration.

3.1 TECHNOLOGIES FOR GROUNDWATER REMEDIATION

The groundwater remedial technologies primarily address the reduction of human health risks posed by the potential use of the groundwater contaminated with VOCs and hexavalent chromium as a domestic drinking water supply. As required by SARA, a range of technologies was identified. These technologies fall into the following seven response categories:

- o no action
- o institutional controls
- o containment on-site
- o collection
- o VOC treatment
- o hexavalent chromium treatment
- o discharge

As the list indicates, a set of separate technologies was identified for the VOCs and hexavalent chromium. The identification of groundwater remedial technologies, including a description of each technology, is shown in Table 3-1. The results of screening, including any site- or waste-limiting characteristics of a technology, are presented in Table 3-2. A summary of the technologies remaining is presented in Table 3-3.

TABLE 3-1
IDENTIFICATION OF
GROUNDWATER REMEDIAL TECHNOLOGIES

CADILLAC FEASIBILITY STUDY

RESPONSE CATEGORY	REMEDIAL TECHNOLOGY	DESCRIPTION OF TECHNOLOGY
No Action	Groundwater Monitoring	Perform water quality analyses to monitor contaminant migration and assess future environmental impacts.
Institutional Controls	Alternate Water Supply	Close existing municipal well system and identify new uncontaminated wells to serve city residents.
Containment On-site	Subsurface Slurry Wall	Emplacement of a low-permeability barrier wall to restrict groundwater migration. Should include a cover system to reduce infiltration.
Collection	Barrier Drain System	A series of shallow trenches constructed to intercept contaminated groundwater.
	Extraction Wells	Installation of several strategically located pumping wells to collect contaminated groundwater for treatment.
VOC Treatment	UV/ozonation	Oxidize organics in extracted groundwater through simultaneous application of ultraviolet light and ozone.
	Air Stripping	Reduce concentrations of volatile organic compounds (VOCs) through intimate contact of extracted groundwater with air. Water descends down a packed column while air is forced up the column to promote mass transfer of organics from aqueous to gaseous phase. Gaseous phase would require further treatment to meet air regulations.
	Steam Stripping	Remove contaminants through intimate contact of extracted groundwater and steam. Similar to air stripping, but steam is utilized to elevate temperatures and enhance removal of volatiles.

TABLE 3-1 (continued)
IDENTIFICATION OF
GROUNDWATER REMEDIAL TECHNOLOGIES

CADILLAC FEASIBILITY STUDY

RESPONSE CATEGORY	REMEDIAL TECHNOLOGY	DESCRIPTION OF TECHNOLOGY
VOC Treatment (continued)		
	Carbon Adsorption	Reduce concentrations of aqueous or gaseous phase organics through adsorption onto available granular activated carbon sites. May be used as a polishing step for treatments such as air stripping to further reduce organic contaminant concentrations.
	Wet Air Oxidation	Destroy organic contaminants in an aqueous solution by inducing oxidation and hydrolysis reactions at high temperature and pressure. Oxygen, at elevated temperatures, enhances oxidation of organic compounds to carbon dioxide and water.
	Thin Film Evaporation	Remove contaminants from extracted groundwater by vaporizing VOCs from water. Process produces a concentrated waste stream requiring further treatment.
	Reverse Osmosis	Remove organics from extracted groundwater using membrane processes. Process will remove organics with >200 molecular weight. Recent studies indicate success in treating organic chemicals with molecular weight >120. At high pressures, membrane allows water to pass while organics are rejected. Process produces a concentrated waste stream requiring further treatment.
	Biological Treatment	Destroy organics through biodegradation, acclimation-degradation, or chemical conversion of the organic wastes by introducing the extracted groundwater to either an aerobic or anaerobic biological treatment process. Microorganisms and nutrients (if needed) are added to induce one or more of the responses.

TABLE 3-1 (continued)
IDENTIFICATION OF
GROUNDWATER REMEDIAL TECHNOLOGIES

CADILLAC FEASIBILITY STUDY

RESPONSE CATEGORY	REMEDIAL TECHNOLOGY	DESCRIPTION OF TECHNOLOGY
VOC Treatment (continued)		
	Supercritical Water Oxidation	Destruction of organic compounds in an aqueous solution at high temperature and pressure. Supercritical water oxidation is a high temperature and pressure version of wet air oxidation.
Hexavalent Chromium Treatment		
	Oxidation-reduction	Chemical reagents are added to contaminated water, changing the oxidation state of the chromium to a more treatable form. This technology would be used prior to precipitation or ion exchange to reduce hexavalent chromium to a less toxic, precipitable trivalent state prior to treatment for removal of the chromium.
	Ion Exchange	Metal ions are removed from solution by exchanging ions, electrostatically attached to a solid resin material, for the dissolved chromium ions in solution. Regeneration of the exhausted resin would produce a concentrated wastestream requiring further treatment.
	Precipitation	Chemical precipitation removes dissolved chromium from solution by chemical reaction. Hexavalent chromium would require reduction to trivalent chromium and precipitate as relatively insoluble hydroxide. Process requires close control of pH. Process would generate chromium hydroxide sludge requiring disposal.

TABLE 3-1 (continued)
IDENTIFICATION OF
GROUNDWATER REMEDIAL TECHNOLOGIES
CADILLAC FEASIBILITY STUDY

RESPONSE CATEGORY	REMEDIAL TECHNOLOGY	DESCRIPTION OF TECHNOLOGY
Hexavalent Chromium Treatment (continued)	Reverse Osmosis	Remove hexavalent chromium from extracted groundwater using membrane processes. Process will remove metals or organics with >200 weight. At high pressures, membrane allows water to pass while organics and metals are rejected. Process produces a concentrated waste stream requiring further treatment.
	Evaporative Recovery	Water is vaporized to concentrate chromium and other metals present in extracted groundwater (e.g. magnesium). Process produces a concentrated waste stream requiring further treatment.
	Electrochemical Reduction	Hexavalent chromium is reduced by an imposed DC electrical potential. Consumable iron electrodes are employed, allowing ferrous ions to be released into solution from the electrodes by the electric current. The ferrous ions reduce the $Cr^{(+6)}$ to $Cr^{(+3)}$. The process produces iron sludge with the precipitated trivalent chromium.
	Activated Carbon Adsorption	Reduce concentrations of hexavalent chromium through adsorption onto available granular activated carbon sites. Process produces spent carbon containing chromium which would require disposal.
Discharge		
	Off-site Water Treatment Facility	Off-site disposal of extracted groundwater to the City of Cadillac publicly owned treatment works (POTW). Groundwater would require transport by means of a force main and/or gravity sewer to the POTW.

TABLE 3-1 (continued)
IDENTIFICATION OF
GROUNDWATER REMEDIAL TECHNOLOGIES

CADILLAC FEASIBILITY STUDY

RESPONSE CATEGORY	REMEDIAL TECHNOLOGY	DESCRIPTION OF TECHNOLOGY
Discharge (continued)		
	Recharge to Groundwater	Reinject treated groundwater using a series of wells and pumps. Can be used to enhance plume removal and accelerate remediation.
	Discharge to Clam River	Discharge treated groundwater directly to nearby surface water body. Transport groundwater by means of a force main.
	Off-site RCRA Disposal Facility	Dispose of extracted groundwater, organic or metals sludges/wastestreams at an off-site RCRA facility. These sludges/wastestreams would be waste residuals from groundwater treatment.

TABLE 3-2

SCREENING OF GROUNDWATER REMEDIAL TECHNOLOGIES

CADILLAC FEASIBILITY STUDY

REMEDIAL TECHNOLOGY	SITE-LIMITING CHARACTERISTICS	WASTE-LIMITING CHARACTERISTICS	SCREENING STATUS
<u>No Action:</u>			
Groundwater Monitoring	This action does not contribute to site remediation.	None.	Retained for further consideration, as required by CERCLA, as amended.
<u>Institutional Controls:</u>			
Alternate Water Supply	This action does not contribute to site remediation. Would need to locate a new well field.	None.	Retain for further consideration as an additional component of a treatment alternative.
<u>Containment:</u>			
Subsurface Slurry Wall	A low-permeability formation is too deep to seat a slurry wall.	None.	Not retained.
Barrier Drain System	Groundwater contamination is too deep and extensive to make this technology effective.	None.	Not retained.
<u>Collection:</u>			
Extraction On-site	None.	None.	Retained for further consideration.
UV/Ozonation	Electrical generation source required for the production of ozone.	Pretreatment for removal of inorganics (e.g., calcium and manganese) present in the groundwater may be required to prevent fouling of ozonation system.	Retained.

TABLE 3-2 (continued)
 SCREENING OF GROUNDWATER REMEDIAL TECHNOLOGIES
 CADILLAC FEASIBILITY STUDY

REMEDIAL TECHNOLOGY	SITE-LIMITING CHARACTERISTICS	WASTE-LIMITING CHARACTERISTICS	SCREENING STATUS
VOC Treatment (continued) UV/Ozonation (continued)		Treatability studies should be performed prior to full-scale design to determine operating parameters and pretreatment requirements necessary to optimize operating efficiency. The contaminant which would control the oxidation retention time and oxidant dose is 1,1,1-trichloroethane.	
Air Stripping	None.	Pretreatment for inorganics such as calcium and magnesium present in the groundwater may be required to prevent fouling of air stripping unit. The contaminant which would control the system design is methylene chloride. Liquid-phase carbon adsorption would be required following air stripping to meet discharge requirements. Off-gases would require treatment by carbon absorption to meet state air regulation.	Retained

TABLE 3-2 (continued)
SCREENING OF GROUNDWATER REMEDIAL TECHNOLOGIES
CADILLAC FEASIBILITY STUDY

REMEDIAL TECHNOLOGY	SITE-LIMITING CHARACTERISTICS	WASTE-LIMITING CHARACTERISTICS	SCREENING STATUS
<u>VOC Treatment (continued)</u>			
Steam Stripping	Electricity, natural gas, or oil source required for steam generation.	Trichloroethylene would be the design compound due to its high concentration. May have to pretreat magnesium magnesium or acidify the feed stream to prevent fouling by the production of compounds such as calcium carbonate on the packing material. Treatability studies should be performed prior to full-scale installation to determine actual system requirements.	Retained.
Carbon Adsorption	None.	Methylene chloride is the least adsorbable of the VOC contaminants of concern and would not be removed to the discharge requirement during aqueous-phase application. Aqueous-phase application would be implemented as polishing step for air stripping to meet discharge requirements. Applicable for vapor-phase application as polishing step for air stripping to meet state air regulations.	Retained for vapor-phase and liquid-phase application following air stripping.

TABLE 3-2 (continued)

SCREENING OF GROUNDWATER REMEDIAL TECHNOLOGIES

CADILLAC FEASIBILITY STUDY

REMEDIAL TECHNOLOGY	SITE-LIMITING CHARACTERISTICS	WASTE-LIMITING CHARACTERISTICS	SCREENING STATUS
<u>VOC Treatment (continued)</u>			
Wet Air Oxidation	On-site mobile treatment systems not available. On-site system would require complete building, resulting in high capital costs. Electricity, natural gas, or oil source required for heat due to low BTU content of groundwater.	This technology is not economical for the dilute (<1 percent) VOC concentrations present in the groundwater due to high treatment costs. Generally achieves about 80 percent oxidation of organic constituents. Therefore, discharge requirements would not be met. This high temperature and pressure system would require extensive monitoring.	Not retained.
Thin Film Evaporation	Electricity, natural gas, or oil source required to produce heat.	Difficult to separate organics with boiling points close to water. Produces a concentrated waste stream requiring further treatment and/or disposal. Not demonstrated at the full-scale level.	Not retained. Process not demonstrated for organics removal.
Reverse Osmosis	None.	Generally applied for dissolved solids and metals removal. Process not proven for lower molecular-weight organics present at the site.	Not retained.

TABLE 3-2 (continued)
SCREENING OF GROUNDWATER REMEDIAL TECHNOLOGIES
CADILLAC FEASIBILITY STUDY

REMEDIAL TECHNOLOGY	SITE-LIMITING CHARACTERISTICS	WASTE-LIMITING CHARACTERISTICS	SCREENING STATUS
<u>VOC Treatment</u> (continued)			
Supercritical Water Oxidation	Electricity, natural gas, or oil source required for heat concen- tration due to low BTU content of VOCs present in groundwater. Full-scale systems not available.	Not economically feasible for dilute concentrations (<1 percent) of VOCs present in the ground- water due to high treatment costs. This high temperature and pressure system would require extensive monitoring.	Not retained.
Biological	None.	Chlorinated organics present in the groundwater are difficult to treat biologically. The VOCs of higher concentrations (e.g., trichloroethylene) would most likely be toxic to the system.	Not retained.
<u>Hexavalent Chromium Treatment:</u>			
Oxidation-Reduction	None.	A reducing agent such as sodium metabisulfite would be introduced to the influent to reduce Cr^{+6} to Cr^{+3} prior to ion exchange.	Retained.
Ion Exchange	None.	Hexavalent chromium (Cr^{+6}) to trivalent chromium (Cr^{+3}) prior to ion exchange.	Retained.

TABLE 3-2 (continued)
 SCREENING OF GROUNDWATER REMEDIAL TECHNOLOGIES
 CADILLAC FEASIBILITY STUDY

REMEDIAL TECHNOLOGY	SITE-LIMITING CHARACTERISTICS	WASTE-LIMITING CHARACTERISTICS	SCREENING STATUS
Hexavalent Chromium Treatment (continued) Ion Exchange (continued)		Regeneration would produce concentrated wastestream requiring further treatment or disposal.	
Precipitation	None.	Would not be able to achieve discharge requirements as a main treatment.	Retain as possible treatment for back-washing effluent produced during ion exchange process.
		Process would generate chromium hydroxide sludge requiring disposal.	
Reverse Osmosis	None.	Difficult to control leakage rate of chromium through membrane. Therefore, may not be able to meet discharge requirements.	
		Would not reduce toxic hexavalent chromium to less-toxic trivalent state. A separation process, not a treatment process. Produces concentrated stream requiring treatment or disposal.	
Evaporative Recovery	Energy source required for evaporation process.	Applicable for process rinse streams to recover chromium for reuse.	Not retained.
		Not cost-effective for chromium levels present in groundwater (<1 ppm).	

TABLE 3-2 (continued)

SCREENING OF GROUNDWATER REMEDIAL TECHNOLOGIES

CADILLAC FEASIBILITY STUDY

REMEDIAL TECHNOLOGY	SITE-LIMITING CHARACTERISTICS	WASTE-LIMITING CHARACTERISTICS	SCREENING STATUS
Hexavalent Chromium Treatment Evaporation Recovery (continued)		VOCs (i.e., 1,1,1-trichloroethane and trichloroethylene) would be released to atmosphere during evaporation process without treatment. May exceed air emissions standards.	
Electrochemical Reduction	Electrical source required to release ferrous ions into sludge solution.	Ferrous ions produce excess (four-fold increase reported) requiring further treatment and/or disposal. The sludge is difficult to settle and dewater due to the presence of ferric hydroxide.	Not retained.
Carbon Adsorption	None.	Treatability testing should be performed to determine actual carbon usage requirements for treatment of hexavalent chromium. Adsorption system should be able to remove VOCs also present in groundwater (trichloroethylene and 1,1,1-trichloroethane). This would provide single-stage treatment for the hexavalent chromium plume. Pretreatment for removal of metals such as calcium and magnesium present in groundwater may be required. Process produces waste residual containing spent carbon, chromium and adsorbed VOCs requiring disposal.	Retained.

TABLE 3-2 (continued)
SCREENING OF GROUNDWATER REMEDIAL TECHNOLOGIES
CADILLAC FEASIBILITY STUDY

REMEDIAL TECHNOLOGY	SITE-LIMITING CHARACTERISTICS	WASTE-LIMITING CHARACTERISTICS	SCREENING STATUS
<u>Disposal:</u>			
Off-site Water Treatment Facility.	Gravity sewer or force main would be required to transport extracted groundwater to POTW.	Capacity of Cadillac POTW would not permit the discharge of an estimated 1.5 million gallons per day (mgd) flow.	Not retained.
Recharge to Groundwater	Appropriate in selected areas where recharge will not affect groundwater flow within capture zones established for extraction of other plumes. Recharge should assist in reducing clean-up times for the Rexair extraction system and possibly the large intermediate aquifer extraction system.	Discharged water must meet applicable discharge requirements for total VOCs and hexavalent chromium.	Retained.
Discharge to the Clam River	Force main or gravity sewer required to transport treated groundwater to discharge point along the Clam River.	Discharged water must meet applicable discharge standards for total VOCs and for total VOCs and hexavalent chromium.	Retained.
Off-site RCRA Disposal Facility	Trucking of extracted groundwater residuals to RCRA facility would be required.	Michigan RCRA facilities accepting metals and organic wastes have been identified. Trucking 1.5 mgd of extracted groundwater is not feasible. Capacity and availability of the RCRA facilities to accept potential waste residuals must be determined.	Retained for disposal of waste residuals. Not retained for disposal of extracted groundwater.

TABLE 3-3
SUMMARY OF SCREENING OF GROUNDWATER REMEDIAL TECHNOLOGIES
CADILLAC AREA FEASIBILITY STUDY

TECHNOLOGY	RETAINED	
	YES	NO
<u>NO-ACTION</u>		
Groundwater Monitoring	X	
<u>CONTAINMENT</u>		
Subsurface Slurry Wall		X
Barrier Drain System		X
<u>EXTRACTION</u>		
Extraction On-site	X	
<u>VOLATILE ORGANIC COMPOUND TREATMENT</u>		
UV/Ozonation	X	
Air Stripping	X	
Steam Stripping	X	
Carbon Adsorption	X	
Wet Air Oxidation		X
Thin Film Evaporation		X
Reverse Osmosis		X
Supercritical Water Oxidation		X
Biological		X
<u>HEXAVALENT CHROMIUM TREATMENT</u>		
Oxidation - Reduction	X	
Ion Exchange	X	
Precipitation	X	
Reverse Osmosis		X
Evaporative Recovery		X
Electrochemical Reduction		X
Carbon Adsorption	X	
<u>DISPOSAL</u>		
Off-site Water Treatment Facility		X
Recharge To Groundwater	X	
Discharge to the Clam River	X	
Off-site RCRA Disposal Facility	X	

3.2 TECHNOLOGIES FOR SOIL REMEDIATION

The purpose of the soil remediation is to prevent continued leaching of VOCs from subsurface soils at Kysor and Four Winns to a potential drinking water source. Soil technologies were categorized into five primary response categories:

- o no action
- o containment
- o on-site disposal
- o off-site disposal
- o treatment

Because the contaminants and site characteristics are similar at Kysor and Four Winns, treatment technologies applicable to contaminants at both areas were identified and are presented in Table 3-4. Treatment technologies were also screened based on their general applicability to both sites. The results of screening are shown in Table 3-5, and the results are summarized in Table 3-6. As soil technologies are combined into remedial alternatives and the alternatives are refined (see Section 5.0), the applicability of an alternative to each site will be discussed.

TABLE 3-4
IDENTIFICATION OF SOIL REMEDIAL TECHNOLOGIES
CADILLAC AREA FEASIBILITY STUDY

RESPONSE CATEGORY	REMEDIAL TECHNNLOGY	DESCRIPTION OF TECHNOLOGY
No Action	Monitoring/Land Use Restrictions	Environmental monitoring and/or deed restrictions.
Containment	Surficial Capping	Diversion of surface water to reduce infiltration of precipitation and subsequent leaching of contaminants into groundwater.
	Subsurface Contain- ment Wall	A vertical subsurface barrier is constructed to control groundwater, and/or leachate movement, in permeable soils. The barrier is constructed of low-permeability material and is usually keyed into a soil confining layer.
Disposal On-site	On-site RCRA Landfill	Disposal of excavated soils in a RCRA-approved landfill constructed on-site.
Disposal Off-site	Off-site RCRA Landfill	Transport and disposal of excavated soils in an off-site RCRA-approved landfill.
Treatment		
(Biological)	In-situ Biodegrada- tion	Microbes are introduced into the soil, or existing soil microbes stimulated, to degrade organic contaminants. The process may require nutrient or oxygen additions to the subsurface environment.
	Biological Treatment	Excavated soil is slurried then introduced into a biological reactor for degradation of organic contaminants.

TABLE 3-4 (continued)
IDENTIFICATION OF SOIL REMEDIAL TECHNOLOGIES

CADILLAC AREA FEASIBILITY STUDY

RESPONSE CATEGORY	REMEDIAL TECHNNNOLOGY	DESCRIPTION OF TECHNOLOGY
(Physical)	Composting	Excavated soils are mixed with organic matter providing a nutrient source for microbes which can then adapt to degrade the organic contaminants.
	Land Spreading	Excavated soil is spread out on the ground surface, and the organic contaminants are reduced through microbial degradation or volatilization.
	Low-Temperature Thermal Stripping	Excavated soil is heated (350 to 700°F) and mixed with air in a process reactor to volatilize organic contaminants.
	Soil Flushing	In-situ soils are flushed with water or other solvent to remove contaminants. The contaminated extract is recovered, then treated with an appropriate water treatment technology, or disposed.
	Solvent Extraction	Excavated soils are mixed with a solvent or water in a batch reactor to extract the contaminant. The extracted solvent is then regenerated or disposed.

TABLE 3-4 (continued)
IDENTIFICATION OF SOIL REMEDIAL TECHNOLOGIES

CADILLAC AREA FEASIBILITY STUDY

RESPONSE CATEGORY	REMEDIAL TECHNNOLOGY	DESCRIPTION OF TECHNOLOGY
(Chemical)	Steam Stripping	Excavated soils are placed in a reactor where steam is added to remove the more water soluble, less volatile contaminants from the soils.
	Vacuum Extraction	Air is circulated through soils in-situ to remove volatile organics. The process gas is recovered and treated to destroy or capture the removed organic contaminants.
	Vitrification	Soil is heated insitu to its melting point using an electrical current to volatilize or combust the organic contaminants.
	Hydrolysis	Excavated soils are slurried with the addition of acid or base to adjust the pH and degrade organic compounds into simpler compounds through a hydrolysis reaction.
	Oxidation - Hydrogen Peroxide	Hydrogen peroxide is injected into soils in situ to oxidize organic contaminants.
(Thermal)	Off-site RCRA Incinerator	Excavated soils are transported to an off-site incineration facility for treatment and disposal.

TABLE 3-4 (continued)
IDENTIFICATION OF SOIL REMEDIAL TECHNOLOGIES

CADILLAC AREA FEASIBILITY STUDY

RESPONSE CATEGORY	REMEDIAL TECHNNLOGY	DESCRIPTION OF TECHNOLOGY
	Fluidized Bed Reactors	Excavated soils are fed into a bed of granular material suspended in hot gases to combust or volatilize organic contaminants.
	In-situ Thermal Destruction	Radio frequency electrodes are placed along the ground surface to heat the shallow subsurface. Organics are destroyed or removed in the super-heated steam resulting from groundwater.
	Infrared Incineration	Excavated soils are fed along a moving conveyor under electric heating elements. Organic contaminants are volatilized from the soil matrix.
	Multiple Hearth Incineration	Excavated soils are fed into the top of a vertical combustion chamber with a series of stacked hearths where the soil goes through drying, combustion, and cooling.
	Pyrolysis Process	Organic constituents are heated in the absence of air to thermally degrade organics into a volatile gaseous portion. Typical pyrolysis equipment is designed to convert organic wastes into boiler fuel.
	Rotary Kiln Incineration	Excavated soils are fed into a heated rotating combustion chamber which tumbles the waste. Contaminants are removed from the soil matrix are fed into a secondary chamber for additional combustion.

TABLE 3-5
SCREENING OF SOIL REMEDIAL TECHNOLOGIES
CADILLAC AREA FEASIBILITY STUDY

REMEDIAL TECHNOLOGY	SITE-LIMITING CHARACTERISTICS	WASTE-LIMITING CHARACTERISTICS	SCREENING STATUS
<u>No Action</u>			
Monitoring/Land Use Restrictions	None.	None.	Retain for further consideration.
<u>Containment</u>			
Surficial Capping	None.	None.	Retain for further consideration.
Subsurface Containment Wall	Confining layer located 60 to 90 feet below grade.	High concentrations of organics can affect the integrity of the containment wall.	Retain for further consideration.
<u>Disposal</u>			
On-site RCRA Landfill	None.	Land disposal regulations.	Retain for further consideration.
Off-site RCRA Landfill	Suitable facility would need to be located.	Land disposal regulations.	Retain for further consideration.
<u>Treatment</u>			
In-situ Biodegradation	None.	Specific organic compounds may be marginally biodegradable.	Retain for further consideration.
Biological Treatment	None.	Specific organic compounds may be marginally biodegradable. The slurring of sandy soils is not considered practical.	Eliminate from further consideration.
Composting	None.	Specific organic compounds may be marginally biodegradable. The relatively large soil volume limits practicability.	Eliminate from further consideration.

TABLE 3-5 (continued)
SCREENING OF SOIL REMEDIAL TECHNOLOGIES
CADILLAC AREA FEASIBILITY STUDY

REMEDIAL TECHNOLOGY	SITE-LIMITING CHARACTERISTICS	WASTE-LIMITING CHARACTERISTICS	SCREENING STATUS
Land Spreading	Limited on-site space.	Specific organic compounds may be marginally biodegradable.	Eliminate from further consideration.
Low-temperature Thermal Stripping	None.	None..	Retain for further consideration.
Soil Flushing	None.	None.	Retain for further consideration.
Solvent Extraction	None.	None.	Retain for further consideration.
Steam Stripping	None.	Technology not sufficiently developed for field implementation.	Eliminate from further consideration.
Vacuum Extraction	None.	None.	Retain for further consideration.
Vitrification	Proximity of treatment system to existing building structures and underground utilities restricts use.	None.	Eliminate from further consideration.
Hydrolysis	None.	Low concentrations of organic contaminants. This process may mobilize naturally-occurring heavy metals.	Eliminate from further consideration.

TABLE 3-5 (continued)
SCREENING OF SOIL REMEDIAL TECHNOLOGIES
CADILLAC AREA FEASIBILITY STUDY

REMEDIAL TECHNOLOGY	SITE-LIMITING CHARACTERISTICS	WASTE-LIMITING CHARACTERISTICS	SCREENING STATUS
Oxidation - Hydrogen Peroxide	None.	Low concentrations of organic contaminants.	Eliminate from further consideration.
Off-site RCRA Incineration	Suitable facility would need to be located.	None.	Retain for further consideration.
Fluidized Bed Reactors	None.	Suitable for on-site incineration.	Retain for further consideration.
In-situ Thermal Destruction	Proximity of treatment system to existing building structures and underground utilities may restrict use.	Conceptual in nature.	Eliminate from further consideration.
Infrared Incineration	None.	Suitable for on-site incineration.	Retain for further consideration.
Multiple Hearth Incineration	None.	Has not been extensively used for hazardous waste destruction.	Eliminate from further consideration.
Pyrolysis	None.	Low concentrations of organic contaminants in soil makes the use of this technology infeasible.	Eliminate from further consideration.
Rotary Kiln Incineration	None.	Suitable for on-site incineration.	Retain for further consideration.

TABLE 3-6
SUMMARY OF SCREENING OF SOIL REMEDIAL TECHNOLOGIES
CADILLAC AREA FEASIBILITY STUDY

TECHNOLOGY	RETAINED	
	YES	NO
<u>NO ACTION</u>		
Monitoring/Land Use Restrictions	X	
<u>CONTAINMENT</u>		
Surficial Capping	X	
Subsurface Containment Wall	X	
<u>DISPOSAL</u>		
On-site RCRA Landfill	X	
Off-site RCRA Landfill	X	
<u>TREATMENT</u>		
In-situ Biodegradation	X	
Biological Treatment		X
Composting		X
Land Spreading		X
Infrared Incineration	X	
Multiple Hearth Incineration		X
Pyrolosis		X
Rotary Kiln Incineration	X	

4.0 DEVELOPMENT AND SCREENING OF GROUNDWATER ALTERNATIVES

This section presents the development and screening of groundwater remedial alternatives. Potential action-specific ARARs, which help to shape remedial alternatives, are presented in Section 4.1. In Section 4.2, these ARARs and the groundwater technologies are used to develop alternatives. In Section 4.3, the alternatives are screened; those retained after screening are presented in Section 4.4.

4.1 POTENTIAL ACTION-SPECIFIC ARARS FOR GROUNDWATER REMEDIAL ALTERNATIVES

Potential action-specific ARARs were identified for the Cadillac site in Section 4.3 of the Cadillac RI Document. This section expands upon those action-specific ARARs for the development of remedial alternatives. Action-specific requirements generally set performance or design standards, controls, or restrictions on particular types of activities. In order to develop technically feasible alternatives, it is important that applicable performance or design standards be considered during the development of all remedial alternatives. Furthermore, selected remedial alternatives are required to comply with all ARARs unless USEPA waives a requirement under one of six conditions identified in Section 4.3 of the RI.

The following paragraphs identify ARARs that will apply to, and must be incorporated in, the proposed remedial alternatives. Based on the technologies that passed initial screening, the proposed alternatives will involve the following

activities: no action; institutional controls; extraction; treatment; and discharges to the Clam River or to the groundwater.

RCRA (40 CFR 264) and Michigan (R 299.9600) regulations for owners and operators of hazardous facilities contain general facility requirements that should be undertaken during all remedial activities, including general waste analysis, security measures (i.e., fencing and posting notices), safety equipment and spill control, emergency planning, and closure and post-closure care. Closure includes monitoring, deed notification, and proper decontamination. Post-closure includes groundwater monitoring and maintenance for 30 years. Depending upon the effectiveness of the remedial alternative in reducing the volume, toxicity, or mobility of, or exposure to, the contamination, the post-closure care period including monitoring may be reduced.

Occupational Safety and Health Administration (OSHA) standards must be recognized and complied with throughout all on-site activities. 29 CFR 1910 contains standards for hazardous waste site operations. Appropriate safety equipment must be located on-site. Allowable levels of contaminants in the atmosphere must be defined and monitored for. If allowable levels are exceeded proper respiratory equipment must be worn.

The construction and operation of on-site treatment facilities must meet design and operation requirements set forth in RCRA and MDNR Hazardous Waste Management and Air Pollution Control regulations. Permits or other administrative requirements for on-site treatment facilities are not required by CERCLA, as

amended. However, because portions of the study area are not considered Superfund/National Priority List (NPL) sites, MDNR may require licenses and permits for the installation and operation of on-site treatment facilities.

Residues from on-site treatment operations must be analyzed and properly handled. If a residue (e.g., spent carbon from carbon absorption) is determined to be a hazardous waste, contingencies must be made for additional treatment and/or the removal and proper disposal. The discharge of treated groundwater, resulting from on-site treatment operations, to the Clam River, will require state approval and an NPDES permit pursuant to the Michigan Water Resources Commission rules (R 323 Parts 1 and 21). NPDES surface water discharge limits will be developed for the Clam River in accordance with the Michigan Water Quality Standards (R 323.1041). The injection of treated wastewater into the ground must have state approval and be in compliance with RCRA underground injection control regulations at 40 CFR 265 (Subpart R) and 40 CFR 144, and the proposed MGQR (R 323 Part 22).

Off-site treatment and/or disposal must be conducted in a permitted facility in compliance with appropriate federal and state hazardous waste facility regulations. Off-site transport of hazardous waste or substances must meet MDNR hazardous waste transportation requirements (R 299.9401). The transporter and vehicle must be licensed with the state.

Table 4-1 identifies the action-specific ARARs for groundwater remediation and details specific requirements.

TABLE 4-1
ACTION-SPECIFIC ARARS
FOR GROUNDWATER
REMEDIALATION ALTERNATIVES

CADILLAC FEASIBILITY STUDY

ARARS	REQUIREMENT SYNOPSIS
<u>Federal</u>	
1. CWA NPDES Program (40 CFR 122, 125)	Point-source discharges from on-site treatment operations must meet NPDES permitting requirements, which include compliance with applicable water quality standards, establishing a discharge monitoring system, and routine completion of discharge monitoring records.
2. RCRA and SDWA UIC Regulations (40 CFR 265 [Subpart R] and 40 CFR 144)	The injection of treated wastewater into the ground must be permitted. Underground injection wells must be constructed, operated, and closed pursuant to RCRA and SDWA regulations.
3. RCRA Location Standards (40 CFR 264.18)	For alternatives incorporating an on-site treatment system, the treatment system must be constructed to withstand a 100-year flood and may not be located within 200 feet of a fault which has had displacement in Holocene time.
4. RCRA Transportation Regulations (40 CFR 262) and DOT Standards (40 CFR Parts 107, 171)	If residue from on-site treatment operations is considered a hazardous waste after treatment, the waste must be transported as a manifested hazardous waste to a permitted RCRA facility. The DOT standards specify packaging, labeling, manifesting, and transporting procedures.
5. RCRA Subtitle D Solid Waste Regulations (40 CFR 241 and 257)	Treatment residues determined to be nonhazardous may be disposed of in a RCRA Subtitle D facility.
6. CAA - NAAQS for Total Suspended Particulates (40 CFR 129.105, 750)	Site remediation activities, including treatment operations, must comply with the (NAAQS). The most relevant pollutant standard is for total suspended particulates at 260 ug/m ³ (primary standard) and 150 ug/m ³ (secondary standard).

TABLE 4-1 (continued)
ACTION-SPECIFIC ARARs
FOR GROUNDWATER
REMEDIALTION ALTERNATIVES

CADILLAC FEASIBILITY STUDY

ARARS	REQUIREMENT SYNOPSIS
State	
7. Michigan Wastewater Discharge Permits (R 323 Part 21)	Discharge to surface waters must be permitted (NPDES) by the state and must comply with monitoring recordkeeping, and reporting requirements. Discharges may not degrade the quality of the water body. MDNR has established a NPDES technology-based level for total VOCs discharged to the Clam River at 20 ppb. The MDNR established a preliminary standard for hexavalent chromium at 6 ppb and total chromium at 73 ppb. Wastewater discharge to a POTW must meet its pretreatment program.
8. Michigan Groundwater Quality (R 232 Part 22)	These regulations are proposed and currently not legally enforceable. Part 22 establishes a nondegradation policy and regulates groundwater discharges. A hydrogeologic report, groundwater monitoring, and a discharge permit are required for all discharges.
9. Michigan Liquid Industrial Waste Disposal Act (MCL c 323)	This act requires that detailed records be maintained and a state-licensed transport vehicle be used for all required industrial waste removals from a facility. Trip records must be maintained, tanks must be tightly closed, and vehicles must be clean. Industrial liquid waste may only be disposed of in a manner and location approved by the commission.
10. Michigan Hazardous Waste Management Regulations (MHWMR), Facility Location Standards (R 299.9603)	Hazardous waste treatment, storage, or disposal facilities may not be located 61 meters of a fault which has had displacement in Holocene time, in a floodway or floodplain, over a sole source aquifer, or in a wetland or coastal high risk area. Active portions of landfills may not be located within 150 meters of commercial, residential, or recreational property lines.
11. Facility Design and Operating Standards (R 299.9604)	Facilities must have runoff controls that prevent flow from a peak discharge of a 24-hour, 25-year storm. Runoff management must control water volume from a 24-hour, 100-year storm.

TABLE 4-1 (continued)
ACTION-SPECIFIC ARARs
FOR GROUNDWATER
REMEDIALATION ALTERNATIVES

CADILLAC FEASIBILITY STUDY

ARARS	REQUIREMENT SYNOPSIS
<u>State (continued)</u>	
12. MHWMR, Construction Permits and Operating Licenses (R 299 Part 5)	A permit to construct and a license to operate a hazardous waste treatment, storage, or disposal facility are required. Application contents, fact sheet requirements, and a permit fee schedule are specified.
13. MHWMR, Transporters of Hazardous Waste (R 299 Part 4)	Requirements go beyond the scope of RCRA transport requirements. A USEPA ID number and Michigan manifest form must be used. Vehicles and drivers are required to have licenses with the state as well as liability insurance. Requirements also specify procedures for proper packaging, identification, and handling of hazardous waste.
14. Michigan Air Pollution Control Regulations (R 366 Parts 2, 3, 7, and 9)	Process or control equipment which may be a source of air contamination is required to have a permit to install and a permit to operate under Part 2. Fugitive dust from trucks, roads, and stockpiles is regulated under Part 3. Part 7 requires that the maximum allowable emission rate from new sources of VOCs be based on the application of BACT. Furthermore, Part 9 prohibits the emission of air contaminants in quantities that will cause "injurious effects to human health or safety, animal life, plant life of significant economic value, or property," or "unreasonable interference with the comfortable enjoyment of life and property."

NOTES:

BACT = best available control technology.
CAA = Clean Air Act.
CWA = Clean Water Act.
DOT = Department of Transportation.
NAAQS = National Air Quality Standards.
NPDES = National Pollutant Discharge Elimination System.
RCRA = Resource Conservation and Recovery Act.

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4.2 DEVELOPMENT OF GROUNDWATER ALTERNATIVES

Groundwater alternatives were developed by assembling the technologies and discharge options that passed the initial screening, and assessing any restrictions imposed by action-specific ARARs. As required by SARA, a range of alternatives was developed to the extent possible.

The groundwater alternatives developed for the Cadillac site include the no-action alternative, and three treatment alternatives that combine extraction, treatment, and discharge. A containment alternative was not developed, because no containment technologies were retained for further consideration. Off-site treatment was also not retained in technology screening except for treatment of residues.

To develop alternatives, the Northernnaire, Shallow, Intermediate, and Rexair Plumes were considered one area to be remediated by one alternative. If it is determined that remediation of the two small VOC plumes in the North Park Subdivision is required, these plumes would then be extracted and the same treatment alternatives used for the other four plumes would be utilized.

4.2.1 No-action Alternative

The no-action alternative consists solely of groundwater monitoring for changes in contaminant concentrations with time along the pathway of migration. No action may be chosen for the entire Cadillac site, or may only be chosen for

the two plumes in the North Park Subdivision, while the other four plumes would be treated.

4.2.2 Pump, Treat, and Discharge Alternatives

Additional components have been identified for the treatment alternatives, GW-2 through GW-4. These include two hexavalent chromium treatment technologies and an alternate water supply option. Hexavalent chromium treatment is associated with each groundwater treatment alternative; therefore, the two technologies will be evaluated against one another during detailed analysis for inclusion in the overall alternative. An alternate water supply is an option that will be evaluated during detailed analysis. The following subsections describe the additional components in greater detail.

The treatment alternatives combine extraction, hexavalent chromium treatment, VOC treatment, and a discharge option. One extraction system has been designed that will capture the four plumes (i.e., Northernnaire, Shallow, Intermediate, and Rexair Plumes). If it is determined that remediation of the VOC plumes in the North Park Subdivision is required, a separate extraction system will be utilized to capture those plumes. The same extraction system will be utilized for Alternatives GW-2 through GW-4.

Each of Alternatives GW-2 through GW-4 consists of a unique combination of a VOC treatment technology and a discharge option. Each VOC technology was combined with either recharge to the groundwater system or discharge to the Clam River. However, in the detailed analysis of alternatives, the treatment

efficiencies of the proposed technologies will be compared to the discharge requirements, and the option to interchange treatment technologies with disposal options will be discussed. A hexavalent chromium treatment technology is also a component of each alternative. This component and any additional components that may be combined with the groundwater alternatives are discussed below.

4.2.3 Additional Components

4.2.3.1 Hexavalent Chromium Treatment. Either of the two hexavalent chromium treatment technologies that passed the initial screening may be a component of Alternatives GW-2 through GW-4. Groundwater extracted from the Northernnaire Plume will first be treated by one of the hexavalent chromium technologies and then discharged to a VOC facility identified in GW-2 through GW-4, for further treatment.

4.2.3.2 Alternate Water Supply. An alternate water supply, which is considered an institutional control, may also be an additional component of Alternatives GW-2 through GW-4. While providing an alternate water supply does not remediate the groundwater contamination, it does eliminate the exposure to contaminated and potentially contaminated drinking water supplies. This institutional control is being considered as an additional component to the groundwater alternatives because total remediation of site groundwater is uncertain and may take many years. Site conditions and past waste disposal practices at the Cadillac site suggest that dense non-aqueous phase liquid (DNAPL) chemicals may be present in the subsurface in the form of residual con-

centrations retained in the soil column or as pools of pure product at the the aquifer base. If DNAPL chemicals are present at Cadillac, groundwater contamination cannot be eliminated in the long-term without removal of the DNAPL source. In most situations, removal of the deep subsurface DNAPL source(s) will not be possible, because there are no highly effective methods presently available for location and removal. The implication is for possible pumping and treatment for DNAPLs in groundwater for an extremely long period of time as the residual sources continue to dissolve into the groundwater. Remedial activities at Cadillac will need to be continually assessed to determine any presence of residual DNAPL chemicals. Providing an alternate water supply may be included with any of the groundwater remedial alternatives being considered for the Cadillac site if it is determined that pumping and treating the groundwater will not remediate the site within the expected time frame.

The groundwater alternatives developed for the Cadillac site and a description of the key components are presented in Table 4-2.

4.3 INITIAL SCREENING OF GROUNDWATER ALTERNATIVES

In this section, the four groundwater alternatives and the three additional components to the alternatives are screened against effectiveness, implementability, and cost factors. Criteria considered for each factor during screening are presented in Table 4-3. In addition, preferences stipulated by SARA with regard to site remediation alternatives are considered when evaluating remedial alternatives. These include the following:

TABLE 4-2
DEVELOPMENT OF GROUNDWATER REMEDIAL ALTERNATIVES
CADILLAC FEASIBILITY STUDY

ALTERNATIVE	DESCRIPTION OF KEY COMPONENTS
GW-1: No Action	- Groundwater monitoring for indicators of contaminant migration.
GW-2: Extraction, Hexavalent Chromium Treatment, UV/Ozonation, Recharge	- Extract contaminated groundwater using pumping well systems. - Hexavalent chromium treatment. - UV/Ozonation for volatile organic compound (VOC) removal. - Discharge treated water back to groundwater system. - Groundwater monitoring for indicators of contaminant migration. - Five-year site review.
GW-3: Extraction, Hexavalent Chromium Treatment, Air Stripping, Discharge	- Extract contaminated groundwater using pumping well systems. - Hexavalent chromium treatment. - Air strip for VOC removal. - Vapor-phase carbon adsorption to meet state air requirements. - Liquid-phase carbon adsorption to meet discharge requirements. - Dispose of spent carbon at off-site RCRA facility. - Discharge treated groundwater to Clam River. - Groundwater monitoring for indicators of contaminant migration. - Five-year site review.
GW-4: Extraction, Hexavalent Chromium Treatment, Steam Stripping, Discharge	- Extract contaminated groundwater using pumping well systems. - Hexavalent chromium treatment. - Steam strip for VOC removal. - Dispose of concentrated waste stream at off-site RCRA facility. - Discharge treated groundwater to Clam River. - Groundwater monitoring for indicators of contaminant migration. - Five-year site review.

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TABLE 4-2 (continued)
DEVELOPMENT OF GROUNDWATER REMEDIAL ALTERNATIVES
CADILLAC FEASIBILITY STUDY

ALTERNATIVE	DESCRIPTION OF KEY COMPONENTS
<u>Additional Components</u>	
AC-1: Hexavalent Chromium Ion Exchange	- Reduction/Ion Exchange for hexavalent chromium removal. - Dispose of regeneration effluent at off-site RCRA facility. - Discharge effluent to VOC facility for further treatment (GW-2, GW-3, or GW-4).
AC-2: Hexavalent Chromium Carbon Adsorption	- Carbon adsorption for hexavalent chromium removal. - Dispose of spent carbon at off-site RCRA facility. - Discharge effluent to VOC facility for further treatment (GW-2, GW-3, or GW-4).
AC-3: Alternate Water Supply	- Restrict access to existing public water supply. - Work with the City of Cadillac to locate and test alternate sources. - Michigan Department of Public Health and citizen approval of new water supply source. - Design and construct new well field and service connections. - Environmental monitoring of water supply. - Conduct educational programs. - Supply bottled water in the interim (if necessary).

TABLE 4-3
 CRITERIA FOR INITIAL SCREENING OF ALTERNATIVES
 CADILLAC AREA GROUNDWATER CONTAMINATION
 CADILLAC FEASIBILITY STUDY

EFFECTIVENESS	IMPLEMENTATION	COST
o Compliance with ARARs	o Technical Feasibility	o Equipment/Construction
o Reduction in Mobility, Toxicity, or Volume of Contaminants	o Demonstrated Performance	o Operation and Maintenance
o Protection of Human Health and the Environment	o Availability of Equipment, Capacity, and Services	
o Adverse Effects Caused by Implementation		

- o Remedial actions that involve treatment to permanently and significantly reduce the volume, toxicity, or mobility of hazardous substances are preferred over remedial actions not involving such treatment.
- o Off-site transport and disposal of hazardous substances without treatment is considered the least-favored alternative when practical treatment technologies are available.
- o Remedial actions utilizing permanent solutions, alternative treatment technologies, or resource recovery technologies are preferred.

Innovative technologies should be carried through screening if there is reason to believe that they offer potential for better treatment performance or implementability, few or lesser adverse impacts than other available approaches, or lower costs than demonstrated technologies. The results of the initial screening are presented in the following subsections.

4.3.1 Alternative GW-1: No-action

This alternative consists solely of groundwater monitoring.

<u>Effectiveness</u>	<u>Implementation</u>	<u>Cost</u>
<u>Advantages</u>	<u>Advantages</u>	<u>Advantages</u>
o Site would be monitored for contaminant migration.	o No remedial actions would be implemented.	o Cost includes groundwater monitoring, maintenance and 5-year review only.
<u>Disadvantages</u>	<u>Disadvantages</u>	<u>Disadvantages</u>
o Would not protect human health from risks due to potential use of VOC- and hexavalent chromium-contaminated groundwater.	o Long-term monitoring and maintenance would be required.	o None.
o Would not reduce the toxicity, mobility, or volume of VOCs and hexavalent chromium in the groundwater.		

Conclusion. The no-action alternative is retained as a baseline for the remaining alternatives.

4.3.2 Pump, Treat, and Discharge Alternatives

The following subsections present the initial screening of pump, treat, and discharge alternatives. These alternatives apply to the Northernair, Rexair, shallow, and intermediate plumes; and apply to the two small VOC plumes in the North Park subdivision if it is determined that remediation is required in that area.

4.3.2.1 Alternative GW-2: Extraction, Hexavalent Chromium Treatment, UV/Ozonation, and Recharge. This alternative consists of groundwater extraction, hexavalent chromium treatment, UV/ozonation for VOC removal, recharging treated water to groundwater system, long-term groundwater monitoring, and a five-year site review.

<u>Effectiveness</u>	<u>Implementation</u>	<u>Cost</u>
<u>Advantages</u>	<u>Advantages</u>	<u>Advantages</u>
- o Would reduce existing risk to human health due to potential use of VOC-contaminated groundwater. - o Would reduce existing risk to human health due to potential use of hexavalent chromium-contaminated groundwater. - o Would reduce toxicity of groundwater by removing contaminants. - o Would not produce secondary wastestream requiring disposal.	- o UV/Ozonation to treat the VOC-contaminated groundwater is technically feasible. - o Equipment and personnel are readily available for on-site implementation.	o Carbon adsorption could be used as a polishing step to reduce capital and operating costs.
<u>Disadvantages</u>	<u>Disadvantages</u>	<u>Disadvantages</u>
- o Hexavalent chromium treatment would produce secondary waste requiring off-site disposal. - o Treatability studies should be performed to determine system requirements.	- o Long-term monitoring and maintenance of treatment system would be required. - o Electrical energy source required for production of ozone.	- o Capital costs for UV/ozonation system estimated to be \$2.2 million. - o Annual operation and maintenance costs estimated at \$1.75 to \$2.00 per 1,000 gallons (approximately \$958,000 \$1 million).

Conclusion. This alternative is retained for detailed analysis because it would reduce existing human health risks due to potential use of contaminated groundwater, it is an innovative technology; and it would not produce a secondary wastestream.

4.3.2.2 Alternative GW-3: Extraction, Hexavalent Chromium Treatment, Air Stripping, and Discharge. This alternative consists of groundwater extraction, hexavalent chromium treatment, air stripping for VOC removal, vapor-phase carbon adsorption to meet state air regulations, liquid-phase carbon adsorption to meet discharge requirements, disposal of spent carbon at off-site RCRA facility, discharging treated groundwater to the Clam River, groundwater monitoring, and a five-year site review.

Effectiveness	Implementation	Cost
<u>Advantages</u>	<u>Advantages</u>	<u>Advantages</u>
o Would reduce existing risk to human health due to potential use of VOC-contaminated groundwater.	o Installation of air stripping and carbon adsorption system at the site is technically feasible.	o Capital costs for the treatment system are estimated at \$730,000.
o Would reduce existing risk to human health due to potential use of hexavalent chromium-contaminated groundwater.	o Full-scale on-site systems are readily available.	o Long-term operation cost would be reduced through on-site regeneration of vapor-phase carbon.
o Would reduce toxicity of groundwater by removing contaminants.		
<u>Disadvantages</u>	<u>Disadvantages</u>	<u>Disadvantages</u>
o Would require off-gas vapor-phase carbon treatment to meet state air regulations.	o Long-term monitoring and maintenance of treatment system would be required.	o Annual operation and maintenance costs estimated at \$567,000.
o Would require liquid-phase carbon treatment to meet discharge requirements.		o Significant annual cost would also be incurred for transport and disposal of spent carbon from liquid-phase application.
o Would require off-site disposal of spent carbon.		
o Hexavalent chromium treatment would produce secondary waste requiring off-site disposal.		
o Pilot-scale studies should be performed prior to full-scale design.		

Conclusion. This alternative is retained for detailed analysis because it would reduce existing human health risks due to potential use of contaminated groundwater, and would have a similar capital cost as the steam stripping alternative.

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4.3.2.3 Alternative GW-4: Extraction, Hexavalent Chromium Treatment, Steam Stripping, and Discharge. This alternative consists of groundwater extraction, hexavalent chromium treatment, steam stripping for VOC removal, disposal of concentrated wastestream at off-site RCRA facility, discharging treated groundwater to Clam River, groundwater monitoring, and a site review every five years.

Effectiveness	Implementation	Cost
<u>Advantages</u> - o Would reduce existing risk to human health due to potential use of VOC-contaminated groundwater. - o Would reduce existing risk to human health due to potential use of hexavalent chromium-contaminated groundwater. - o Would reduce toxicity of groundwater contaminants. - o Would not produce air emissions. <u>Disadvantages</u> - o Would require disposal of condensate at off-site RCRA facility. - o Hexavalent chromium treatment would produce secondary waste requiring off-site disposal. - o Treatability studies should be performed to determine system requirements.	<u>Advantages</u> - o Installation of a steam stripping system is technically feasible for the groundwater VOC contaminants at the site. - o Steam stripping has been demonstrated for groundwater treatment. - o Process equipment is readily available. <u>Disadvantages</u> - o Not as readily implementable as air stripping system because of greater equipment requirements and time required to tie into electrical energy source. - o Long-term monitoring and maintenance would be required.	<u>Advantages</u> - o Capital costs (for steam stripping system equipment including reboiler for vapor recompression) are estimated to be \$700,000. <u>Disadvantages</u> - o Long-term operating costs associated with generation of process steam could be reduced through use of vapor recompression in the system, versus a straight steam process (approximately \$185,000 versus \$500,000 per year). - o Long-term annual costs associated with disposal of waste condensate.

Conclusion. This alternative is retained for detailed analysis because it would reduce existing human health risks due to potential use of contaminated groundwater, and would not produce air emissions, and has a similar capital cost to that of the air-stripping system.

4.3.3 Additional Components

This section presents the initial screening of components that may be added to the groundwater alternatives screened in the previous subsection.

4.3.3.1 Additional Component AC-1: Hexavalent Chromium - Ion Exchange. This additional component consists of ion exchange for hexavalent chromium removal, disposal of regeneration effluent and spent resin at an off-site RCRA facility, and discharging effluent to a VOC facility for further treatment (GW-2, GW-3, or GW-4).

Effectiveness	Implementation	Cost
<u>Advantages</u>	<u>Advantages</u>	<u>Advantages</u>
- o Would reduce existing risk to human health due to potential use of hexavalent chromium-contaminated groundwater. - o Would reduce toxicity of groundwater by removing contaminants.	- o Installation of an ion exchange system at the site is technically feasible. - o Equipment is readily available and requires minimal time for full-scale installation.	- o Relatively low capital cost for ion exchange system (approximately \$55,000). - o Precipitation may be added to reduce the quantity of regeneration effluent requiring disposal.
<u>Disadvantages</u>	<u>Disadvantages</u>	<u>Disadvantages</u>
o Would require disposal of regeneration effluent and spent resin at off-site RCRA facility.	o Long-term monitoring and maintenance of treatment system would be required.	- o Long-term costs associated with operation and maintenance of the system. - o Long-term costs associated with disposal of regeneration effluent.

Conclusion. This additional component is retained for detailed analysis because it would reduce existing human health risks due to potential use of contaminated groundwater, it would be easy to implement, and it has a low capital cost.

4.3.3.2 Additional Component AC-2: Hexavalent Chromium - Carbon Adsorption. This additional component consists of carbon adsorption for hexavalent chromium removal, disposal of spent carbon at an off-site RCRA facility, and discharging effluent to Clam River.

Effectiveness	Implementation	Cost
<u>Advantages</u> - o Would reduce existing risk to human health due to potential use of hexavalent chromium-contaminated groundwater. - o Could reduce existing risk to human health due to potential use of contaminated groundwater by also treating the VOCs contained in this plume.	<u>Advantages</u> - o Installation of a carbon adsorption system at the site is technically feasible. - o Equipment is readily available and relatively easy to implement.	<u>Advantages</u> - o Capital costs for carbon adsorption system estimated at \$60,000. - o Single-stage treatment could reduce overall costs by reducing flow to main VOC treatment system.
<u>Disadvantages</u> - o Would require disposal of spent carbon at an off-site RCRA facility. - o Treatability studies should be performed to determine system requirements.	<u>Disadvantages</u> o Long-term monitoring and maintenance of the system would be required.	<u>Disadvantages</u> - o Long-term costs associated with operation and maintenance of the system. - o Long-term costs associated with disposal of spent carbon.

Conclusion. This additional component is retained for detailed analysis because it would reduce existing human health risks due to potential use of contaminated groundwater, could provide single-stage treatment of the Northernnaire plume, and has a similar capital cost as the ion exchange process.

4.3.3.3 Additional Component AC-3: Alternate Water Supply. This additional component involves relocating the well field to provide a new source of public drinking water.

Effectiveness	Implementation	Cost
<u>Advantages</u> o Would eliminate exposure of contaminants in drinking water by utilizing a different water supply source.	<u>Advantages</u> o The City of Cadillac has been advised by the Michigan Department of Public Health (MDPH) to locate an alternate source of drinking water and has located one potential new source. The implementation of this alternative has already begun. The technology and construction will be similar to the existing well field and water supply facility.	<u>Advantages</u>
<u>Disadvantages</u> o Would not remediate existing groundwater contamination. Therefore, this alternative, used by itself, would not meet groundwater quality and safe drinking water ARARs.	<u>Disadvantages</u> o The process of locating an "acceptable" new water supply may take several months to several years.	<u>Disadvantages</u> o High capital costs associated with closing the existing well field and constructing a new well field. Implementation will be labor and equipment-intensive. The long-term costs associated with operation and maintenance will be the same as for the existing well field.

Conclusion. According to conversations with MDPH Division of Water Supply, the City of Cadillac was advised by MDPH to develop a contingency plan for the Cadillac well field (i.e., identify another well field location). It is the policy of MDPH to pursue elimination of any trace of VOC contamination from public water supplies through constructing new wells or alternate water supply sources. Therefore, the institutional control of moving the well field may already be in progress. This alternative will be retained as a potential additional component of another groundwater alternative.

4.4 SUMMARY OF SCREENING OF REMEDIAL ALTERNATIVES FOR GROUNDWATER

The groundwater remedial alternatives retained after the screening process for detailed analysis follow:

- o GW-1: No-action
- o GW-2: Extraction, Hexavalent Chromium Treatment, UV/Ozonation,
Recharge
- o GW-3: Extraction, Hexavalent Chromium Treatment, Air Stripping,
Discharge
- o GW-4: Extraction, Hexavalent Chromium Treatment, Steam Stripping,
Discharge
- o AC-1: Hexavalent Chromium Treatment - Ion Exchange
- o AC-2: Hexavalent Chromium Treatment - Carbon Adsorption
- o AC-3: Alternative Water Supply

5.0 DEVELOPMENT AND SCREENING OF SOIL REMEDIAL ALTERNATIVES

Alternatives applicable to the soil contamination at the Kysor and Four Winns sites are developed and screened in this section. Potential action-specific ARARs that aid in the development of soil alternatives are presented in Section 5.1. The approach and results of the development of soil alternatives are discussed in Section 5.2. In Section 5.3, the alternatives are screened; those retained after screening are summarized in Section 5.4.

5.1 POTENTIAL ACTION-SPECIFIC ARARS FOR SOIL REMEDIAL ALTERNATIVES

Alternatives initially considered for soil remediation include the following activities: no action, excavation, treatment on-site, treatment off-site, in-situ treatment, and capping. Remedial alternatives involving excavation and on-site treatment operations must comply with federal and state air regulations. The federal Clean Air Act regulations contain National Ambient Air Quality Standards (NAAQS) and National Emission Standards for Hazardous Air Pollutants (NESHAPs) which may become applicable. The Michigan Administrative Rules for Air Pollution Control (R 336.1101) contain emissions limitations and prohibitions for emissions that are deemed to have injurious effects on human health and welfare and the environment. Best available control technology (BACT) requirements may also apply to new treatment facilities.

All remedial alternatives must comply with the RCRA closure requirements. For remedial alternatives attempting to clean close, all waste residues,

contaminated containment and treatment system components, and contaminated subsoils must be removed and managed as hazardous waste. After a site is certified as clean closed, no further monitoring or post-closure care is required (see 40 CFR 264.113, .228, and .258). For alternatives that leave wastes or contaminated soils on-site, the site would need to be capped with a final cover designed and constructed to provide long-term minimization of the migration of leachate. This type of closure requires post-closure care for a minimum of 30 years, at the discretion of the Regional Administrator.

The construction and operation of on-site treatment facilities (e.g., an incinerator) must meet the design and operation requirements set forth in RCRA and MDNR Hazardous Waste Management and Air Pollution Control regulations. Permits, licenses, or other administrative requirements for on-site treatment facilities are not required by CERCLA, as amended. However, because only the Northernaire site is currently considered a federal Superfund/NPL site (the Kysor site is currently nominated to be placed on the NPL), it is unclear whether MDNR will require permits for the installation and operation of on-site treatment facilities.

Residues, scrubber water, ash, or atmospheric emissions from any on-site treatment activities must be anticipated and properly handled. For example, ash or other residues must be analyzed to determine if they are considered hazardous waste. If so, contingencies must be made for either the treatment of and/or removal and proper disposal of the waste (e.g., in a permitted facility).

If an alternative involves land disposal without treatment, RCRA "land ban" would be applicable. "Land ban" prohibits the land disposal of hazardous wastes and residuals not meeting specified treatment standards. The Hazardous and Solid Waste Amendments (HSWA) and the RCRA land ban regulations (40 CFR 268) establish categories of hazardous wastes and specify a schedules to ban these wastes from land disposal without treatment. These standards are to be achieved by the best demonstrated available treatment technologies (BDATs). A BDAT may either be a concentration- or technology-based standard.

Certain solvent-contaminated wastes were banned from land disposal on November 8, 1986. Solvent contaminated soil and debris resulting from a response action taken under Section 104 or 106 of CERCLA have been granted an extension until November 8, 1988. The contaminated soils at the Cadillac site fall into this category. Therefore, land disposal of untreated soils and debris from Cadillac will not be feasible after November 8, 1988, unless USEPA has:

- approved a petition demonstrating that hazardous constituents will not migrate from the land disposal unit for as long as the wastes remain hazardous;
- granted an extension to the effective date of the prohibition due to insufficient waste treatment capacity; or
- granted a variance to the treatment standard because the waste is significantly different from that which is regulated under the land ban.

It should be noted that Michigan has adopted the RCRA land ban regulations in R 299.9627.

Atmospheric emissions must be controlled and must comply with the Michigan Administrative Rules for Air Pollution Control. For remedial alternatives incorporating off-site transport of hazardous substances, the transporter and vehicle must be licensed by the state and must comply with Department of Transportation manifest and other MDNR transporter requirements (R 299 9401). Any off-site treatment or disposal facilities must be permitted and in compliance to the appropriate federal and state hazardous waste facility regulations.

Table 5-1 identifies the action-specific ARARs for soil remediation, and provides detail on specific requirements.

5.2 DEVELOPMENT OF SOIL ALTERNATIVES

Based on the applicable technologies and potential action-specific ARARs identified in the preceding sections, eleven soil remedial alternatives were developed for Cadillac site. These alternatives were developed based on their general applicability to both the Kysor and Four Winns contaminated soils. As required by SARA, the range of alternatives include no-action, containment, and treatment. The alternatives developed for Cadillac are presented in Table 5-2 and are briefly described in the following subsections.

TABLE 5-1
ACTION-SPECIFIC ARARS FOR SOIL
REMEDIALATION ALTERNATIVES

CADILLAC FEASIBILITY STUDY

ARARS	REQUIREMENT SYNOPSIS
<u>Federal</u>	
1. RCRA Regulations, 40 CFR 264 Landfill Regulations (40 CFR 264 [Subpart N])	For capping in-place alternatives, a final cover must be designed, constructed at closure to prevent migration of liquids, have minimal maintenance requirements, promote drainage, minimize erosion, and accommodate settling. The cover integrity and the leachate collection, groundwater monitoring and runoff/runon systems must be maintained throughout the post-closure care period.
2. RCRA Location Standards (40 CFR 264.18)	For alternatives incorporating an on-site cap or treatment system, the facility or cap must be constructed to withstand a 100-year flood and may not be located within 200 feet of a fault which has had displacement in Holocene time.
3. RCRA Incinerator Standards (40 CFR 264 [Subpart O])	For alternatives involving on-site incineration, the mobile incinerator must be permitted or approved by USEPA. Waste analyses and a trial burn is required to demonstrate removal efficiency. The on-site incinerator must achieve a destruction and removal efficiency of 99.9 percent of the Principal Organic Hazardous Constituent (POHC). Hydrogen chloride (HCl) stack emissions will be controlled to no greater than the larger of 1.8 kg/hr or 1 percent of the HCl in the stack gas prior to pollution control equipment.
4. RCRA Storage Regulations (40 CFR 264 [Subpart I])	For alternatives specifying incineration, wastes may have to be stored onsite following excavation while awaiting incineration. Wastes must be stored in such a manner as to preclude leaks and withstand flooding.

TABLE 5-1 (continued)
ACTION-SPECIFIC ARARS FOR SOIL
REMEDIAL ALTERNATIVES

CADILLAC FEASIBILITY STUDY

ARARS	REQUIREMENT SYNOPSIS
<u>Federal (continued)</u>	
5. RCRA Transportation Regulations (40 CFR 262) and DOT Standards (40 CFR 107 and 171)	If incinerator ash residue is considered a hazardous waste after treatment, the waste must be transported as a manifested hazardous waste to a permitted RCRA facility. The DOT standards specify packaging, labeling, manifesting, and transporting procedures.
6. RCRA Land Disposal Regulations (40 CFR 268)	Restricts the land disposal of certain wastes without treatment and specifies treatment standards for these wastes. Solvent wastes generated at CERCLA sites are banned from land disposal as of November 8, 1988. Any waste meeting this definition at the Cadillac site will need to be treated prior to land disposal.
7. RCRA Subtitle D Solid Waste Regulations (40 CFR 241 and 257)	Incinerator residues determined to be nonhazardous may be disposed of in a RCRA Subtitle D facility.
8. RCRA Standards for Chemical, Physical, and Biological Treatment (40 CFR 265 Subpart Q)	Solidification, to the extent practical, must follow these general operating requirements. Trial tests and water analyses must be performed. At closure, all waste residues must be removed from the site and equipment must be decontaminated.
9. CAA - NAAQS for Total Suspended Particulates (40 CFR 129.105 and 750)	Site remediation activities must comply with the National Ambient Air Quality Standards (NAAQS). The most relevant pollutant standard is for total suspended particulates, at $260 \mu\text{g}/\text{m}^3$ (primary standard) and $150 \mu\text{g}/\text{m}^3$ (secondary standard).

TABLE 5-1 (continued)
ACTION-SPECIFIC ARARS FOR SOIL
REMEDATION ALTERNATIVES

CADILLAC FEASIBILITY STUDY

ARARS	REQUIREMENT SYNOPSIS
<u>State</u>	
10. Michigan Hazardous Waste Management Regulations (MHWMR) Facility Location Standards (R 299.9603)	Hazardous waste treatment, storage, or disposal facilities may not be located within 200 feet of a fault which has had displacement in the Holocene period, in a floodway or floodplain, over a sole source aquifer, or in a wetland or coastal high risk area. Active portions of landfills may not be located within 492 feet of commercial, residential, or recreational property lines.
11. MHWMR, Facility Design and Operating Standards (R 299.9604)	Runon controls must prevent flow from a peak discharge of a 24-hour, 25-year storm. Runoff management must control water volume from a 24-hour, 100-year storm.
12. MHWMR, Use and Management of Containers (R 299.9614)	Refers to RCRA regulations at 40 CFR 264 (Subpart I).
13. MHWMR, Landfills (R 299.9619)	Refers to 40 CFR 264 (Subpart N) (Landfills). Additional requirements are (1) the final cover must be compacted clay which meets the liner requirements (a unified soil classification system with over 25 percent of the soil particles less than 5 microns in size); (2) not less than 24 inches of additional material is required over the final cover to prevent animal burrowing into the synthetic cover component; (3) establish shallow rooted grass to stabilize cap and prevent erosion at the earliest possible point; and (4) establish a venting system to prevent accumulation of gases.
14. MHWMR, Incinerators (R 299.9623 through .9626)	These standards reference portions of the federal RCRA regulations (40 CFR 264 [Subpart O]) including management of POHCs, incinerator operation standards, and trial burns. Additional requirements specify design criteria including temperature and retention time standards; monitoring of oxygen content; and additional trial operations.

TABLE 5-1 (continued)
ACTION-SPECIFIC ARARS FOR SOIL
REMEDIAL ALTERNATIVES

CADILLAC FEASIBILITY STUDY

ARARS	REQUIREMENT SYNOPSIS
<u>State (continued)</u>	
15. MHWMR, Construction Permits and Operating Licenses (R 299 Part 5)	A permit to construct and a license to operate a hazardous waste treatment, storage, or disposal facility are required. Application contents, fact sheet requirements, and a permit fee schedule are specified.
16. MHWMR, Transporters of Hazardous Waste (R 299 Part 4)	Requirements go beyond the scope of RCRA transport requirements. A USEPA ID number and Michigan manifest form must be used. Vehicles and drivers are required to have licenses with the state as well as liability insurance. Requirements also specify procedures for proper packaging, identification, and handling of hazardous waste.
17. Michigan Air Pollution Control Regulations (R 366 Parts 2, 3, 7, and 9)	Process or control equipment that may be a source of air contamination is required to have a permit to install and a permit to operate under Part 2. Fugitive dust from trucks, roads, and stockpiles is regulated under Part 3. Incinerators with a capacity of 100 pounds or less per hour have a maximum allowable emission of 0.65 pounds of particulate per 1,000 pounds of gas. Incinerators with a capacity over 100 pounds/hour have a maximum allowable emission limit of 0.30. Part 7 requires that the maximum allowable emission rate from new sources of VOCs be based on the application of BACT. Furthermore, Part 9 prohibits the emission of air contaminants in quantities that will cause "injurious effects to human health or safety, animal life, plant life of significant economic value, or property," or "unreasonable interference with the comfortable enjoyment of life and property."

NOTES:

BACT = best available control technology.

CAA = Clean Air Act.

CWA = Clean Water Act.

DOT = Department of Transportation.

NAAQS = National Air Quality Standards.

NPDES = National Pollutant Discharge Elimination System.

RCRA = Resource Conservation and Recovery Act.

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TABLE 5-2
DEVELOPMENT OF SOURCE CONTROL ALTERNATIVES

CADILLAC AREA FEASIBILITY STUDY

ALTERNATIVE	DESCRIPTION OF KEY COMPONENTS
SC-1: No-action	- Groundwater monitoring to indicate any migration of contaminants from the affected soils. - Deed restrictions to notify future owners of the presence of contaminated soils. - Long-term monitoring and maintenance of the site.
SC-2: Surficial Capping/ Subsurface Containment Wall	- Post-closure plan to monitor and maintain site. - Composite cap system to minimize surface water infiltration. - Bentonite slurry wall to prevent groundwater migration. - Groundwater monitoring to indicate any migration of contaminants from the affected soils. - Surface water management to minimize erosion of final cover. - Vegetative establishment to minimize erosion of final cover. - Periodic five-year review of remedial actions.
SC-3: Excavation, Off-site RCRA Landfill	- Excavation of contaminated soil to the water table. - Transport of excavated soil to a RCRA-permitted landfill for disposal.
SC-4: Excavation, On-site RCRA Landfill	- Excavation of contaminated soil to the water table. - Construction of an on-site RCRA landfill to contain excavated soils. - Installation of a groundwater monitoring system meeting RCRA regulations. - Groundwater monitoring to indicate any migration of contaminants from the affected soils. - Long-term monitoring and maintenance of the site. - Periodic five-year review of remedial actions.
SC-5: In-situ Biodegradation	- Introduction or enhancement of waste-degrading microbes in the affected soils. - A nutrient and oxygen source is provided to sustain the microbial activity. - Post-treatment soil sampling for verification of process effectiveness.

TABLE 5-2 (continued)
DEVELOPMENT OF SOURCE CONTROL ALTERNATIVES

CADILLAC AREA FEASIBILITY STUDY

ALTERNATIVE	DESCRIPTION OF KEY COMPONENTS
SC-6: Excavation, Low-temperature Thermal Stripping	- Excavation of contaminated soil to the water table. - Excavated soil is processed in a low-temperature reactor to desorb organic contaminants. - Removed organics are vented to the atmosphere to volatilize, captured in a vapor-phase carbon canister, or thermally destroyed in an afterburner.
SC-7: In-situ Soil Flushing	- A source of water (flushing agent) is required. - Construction of infiltration basin typically consisting of aggregate base and polyethylene distribution pipe. - Subsurface system to collect flushate for subsequent treatment (extraction wells). - Flushate treatment system (groundwater treatment alternative). - Groundwater monitoring during process implementation. - Post-treatment soil sampling for verification of process effectiveness.
SC-8: Solvent Extraction	- Excavation of contaminated soil to the water table. - Process reactor to slurry and mix excavated soils with solvent. - Dewatering device to remove solvent following processing. - Solvent collection for treatment or disposal. - Solvent treatment system if solvent reuse is anticipated.
SC-9: In-situ Vacuum Extraction	- Installation of wells in unsaturated zone to circulate subsurface vapor. - Blower mechanism to induce vapor circulation. - Volatilized organics are vented to the atmosphere to be captured in a vapor-phase carbon canister or thermally destroyed in an afterburner. - Post-treatment soil sampling for verification of process effectiveness.

TABLE 5-2 (continued)
DEVELOPMENT OF SOURCE CONTROL ALTERNATIVES

CADILLAC AREA FEASIBILITY STUDY

ALTERNATIVE	DESCRIPTION OF KEY COMPONENTS
SC-10: Excavation, Off-site Incineration	- Excavation of contaminated soil to the water table. - Transport of excavated soil to a RCRA-permitted off-site incinerator. - Residual process ash must be disposed of.
SC-11: Excavation, On-site Incineration	- Excavation of contaminated soil to the water table. - Processing of excavated soil in an on-site mobile incinerator. - Installation of air pollution control equipment to control emissions from process equipment. - Air monitoring during process operation. - Residual process ash disposed of on-site.

5.2.1 No-action Alternative

The no-action alternative would involve groundwater monitoring to indicate any migration of contaminants from the affected soils. Restrictions would be placed on deeds to notify future owners of the presence of contaminated soils. Under the no-action alternative, the site must be reviewed every five years by USEPA.

5.2.2 Containment, Treatment, and Disposal Alternatives

Three containment alternatives were developed for the Cadillac site. These alternatives address applicable closure and post-closure requirements.

The nine remaining alternatives incorporate treatment as a principal element, but vary in the degree to which they minimize long-term management and reduce toxicity, mobility, or volume of contaminants. In addition, one alternative involves off-site treatment.

5.3 INITIAL SCREENING OF SOIL ALTERNATIVES

Alternatives were screened for applicability to both the Kysor and Four Winns sites. Two options were considered suitable in formatting the presentation of the screening: (1) screen the developed alternatives considering each site individually, thereby requiring 22 alternatives to be addressed in the initial screening; or (2) screen each alternative considering its applicability to the sites in general terms, noting any site-specific differences or requirements.

To determine the best option, the specific characteristics of each site were first analyzed to highlight similarities and differences. Table 5-3 presents the results of this analysis.

Comparing the specific characteristics of each site reveals that the sites are very similar. The estimated volumes of soils to be remediated are relatively close, contaminants of concern at each site can all be categorized as VOCs, and the geology and depth to groundwater at each site are nearly identical. The major differences between the individual sites are the specific VOCs that prompt site remediation, and their identified concentrations (i.e., chemicals in soil at the Kysor site appear in relatively higher concentrations).

Considering the similarities between the two sites, it was decided that it would be most appropriate to screen the alternatives concurrently for both sites. Any differences or requirements specific to each individual site (Kysor versus Four Winns) are noted. The results of the initial screening are presented in the following subsections.

TABLE 5-3
PRELIMINARY SITE EVALUATION
AREAS OF SOIL REMEDIATION

CADILLAC FEASIBILITY STUDY

SITE CHARACTERISTIC	KYSOR SITE	FOUR WINNS SITE
Estimated Surface Area of Contamination	250 ft. by 75 ft. = 18,750 s.f.	270 ft. by 70 ft. = 18,900 s.f.
Estimated Depth to Water Table	25 feet	23 feet
Zone of Contaminated Soil (depth below grade)	6 to 25 feet	0 to 23 feet
Estimated Volume of Soil to be Remediated	13,200 cy	16,100 cy
Chemicals Present which Exceed Target Clean-up Levels (TCLs)	TCE, Xylene, 1,1,1 TCA	MC*
Target Clean-up Levels (ppm)	TCE: 0.07 Xylene: 141 1,1,1-TCA: 7.6	MC: 0.02
Maximum Concentration (ppm)	TCE: 74.00 Xylene: 520 1,1,1-TCA: 24	MC: 4.70
Soil Type (unsaturated zone)	med. to fine sand	med. to fine sand
Site Accessibility	very good	good some gravel/pavement

NOTES: cy = cubic yards; MC = methylene chloride; 1,1,1 TCA = 1,1,1 trichloroethane; TCE = trichloroethylene.

- * Sparse soil data available for Four Winns are not believed to be sufficient for complete characterization. A groundwater plume containing 1,1,1-TCA has been identified as related to Four Winns. A residual concentration of 1,1,1-TCA would also be expected in soils; however, remediation of soils to the TCL for MC may control soil remediation volumes and a conservative distribution of MC has been assumed for the site.

5.3.1 Alternative SC-1: No Action

The no-action alternative consists of groundwater monitoring of the site and deed restrictions to protect public health.

<u>Effectiveness</u>	<u>Implementation</u>	<u>Cost</u>
<u>Advantages</u>	<u>Advantages</u>	<u>Advantages</u>
o Site would be monitored for contaminant migration.	o The installation of additional monitoring wells is technically feasible and easily implementable.	o Cost includes groundwater monitoring, maintenance, and 5-year review only.
<u>Disadvantages</u>	<u>Disadvantages</u>	<u>Disadvantages</u>
o Would not reduce the residual soil contamination currently contributing to the groundwater contamination because chemicals would continue to leach from soil into the groundwater.	o A periodic 5-year review of the site would be required.	o None.
o Would not reduce the mobility, toxicity, or volume of residual soil contaminants.	o Long-term monitoring would be required.	
o Would not remediate a source of groundwater contamination.		
o The effectiveness of deed restrictions is considered marginal.		

Conclusion. The no-action alternative is retained for detailed analysis as a baseline for the remaining alternatives.

5.3.2 Containment, Treatment, and Disposal Alternatives

The following subsections present the initial screening of containment, and treatment and disposal alternatives for the Kysor and the Four Winns sites. Unless specifically noted, comments are applicable to each site.

5.3.2.1 Alternative SC-2: Surficial Capping/Containment Wall. This alternative consists of surficial capping of the site coupled with the use of a subsurface containment wall implementing surface water management, groundwater monitoring, and post-closure plan development.

<u>Effectiveness</u>	<u>Implementation</u>	<u>Cost</u>
<u>Advantages</u>	<u>Advantages</u>	<u>Advantages</u>
o Designed to reduce the infiltration of precipitation through contaminated soils thereby reducing the mobility of contaminants.	- o Installation of a cap system can be easily implemented and is considered technically feasible. - o Does not require the excavation of large volumes of contaminated soil.	
<u>Disadvantages</u>	<u>Disadvantages</u>	<u>Disadvantages</u>
- o Would not reduce the volume or toxicity of the residual soil contamination chemicals of concern. - o Contaminants would remain in on-site soils.	- o Would require long-term monitoring and maintenance. - o Confining layer beneath site is relatively deep (60 to 90 feet), which limits the desirability of installing a subsurface containment wall. - o Would require a periodic five-year review of the site.	o Installation of a deep containment wall would increase cost.

Conclusion. RI/FS guidance suggests a containment alternative be carried through the alternative screening for subsequent detailed evaluation. While this alternative may have some constraints associated with the installation of a subsurface containment wall, it is retained for detailed analysis for use at both the Kysor and Four Winns sites. If during the detailed assessment the subsurface containment wall is proven to be infeasible for implementation at the sites, the alternative will be evaluated based solely on the merits of the cap system.

5.3.2.2 Alternative SC-3: Off-site RCRA Landfill. This alternative consists of the excavation of all soils that exhibit chemical concentrations exceeding the established TCLs, and transport of contaminated soils to a RCRA-permitted landfill for disposal.

Effectiveness	Implementation	Cost
<u>Advantages</u>	<u>Advantages</u>	<u>Advantages</u>
o Would reduce risks associated with residual soil contamination by removing soil.	o Eliminates the need to construct and operate process equipment at the site. o Likely to be the most time-effective alternative.	o Capital short-term costs would be less than on-site landfill alternative.
<u>Disadvantages</u>	<u>Disadvantages</u>	<u>Disadvantages</u>
o Off-site transport and disposal is least preferred alternative per SARA.	o Soil from the Kysor site would be subject to land ban restrictions affecting CERCLA response actions.	o None.
o Potential exposure risk associated with the transport of contaminated soil.	o Soil from the Four Winns site may be subject to land ban restrictions.	
o Toxicity, mobility, and volume of contaminants would be reduced by removing soil from the site, but would be transferred to another site.	o Requires extensive soil excavation.	

Conclusion. Because the soil from the Kysor site would be subject to land ban restrictions, soils from the Four Winns site could be subject to land ban restrictions if they are classified as a RCRA waste and/or if they fail the Toxicity Characteristic Leaching Procedure (TCLP). For this reason, and because disposal without treatment is least-preferred under SARA, this alternative is not retained for detailed analysis.

5.3.2.3 Alternative SC-4: On-site RCRA Landfill. This alternative consists of the construction of a RCRA landfill cell at each of the sites to contain excavated soils that exhibit chemical concentrations exceeding the TCLs, or construction of a RCRA landfill cell at one area to consolidate materials from both sites.

Effectiveness	Implementation	Cost
<u>Advantages</u> o Affected soils are contained in an engineered landfill cell, thereby reducing the potential for leaching mobility.	<u>Advantages</u> o Kysor site would be exempt from obtaining RCRA permit but would have to meet the intent of the regulations.	<u>Advantages</u> o None.
<u>Disadvantages</u> o Toxicity and volume of contaminants would not be reduced.	<u>Disadvantages</u> - o Four Winns site would require a RCRA Part B permit to dispose of wastes on-site. - o Long-term monitoring and maintenance would be required. - o An on-site RCRA landfill would be subject to the same land ban restriction as an off-site facility. - o Would require a periodic five-year review of the site.	<u>Disadvantages</u> - o Considerable cost and manpower are required to construct, maintain, and monitor an on-site facility. - o Long-term costs associated with operation and maintenance of landfill cover system.

Conclusion. As with Alternative SC-3, the soil from the Kysor site would be subject to land ban restrictions, soils from the Four Winns site could be subject to land ban restrictions if they are classified as a RCRA waste and/or if they fail the TCLP, and disposal without treatment is least-preferred under SARA. In addition, Four Winns soils would require a RCRA Part B permit to dispose on-site. Therefore, this alternative is not retained for detailed analysis.

5.3.2.4 Alternative SC-5 In-situ Biodegradation. This alternative consists of in-situ biodegradation of organic chemicals in soils to levels below the established TCLs.

Effectiveness	Implementation	Cost
<u>Advantages</u> - o Process degrades organic contaminants, thereby reducing their toxicity, volume, and mobility. - o Permanent treatment remedy. <u>Disadvantages</u> - o Treatability testing would be required to assess site-specific effectiveness. - o Water, microbes, and nutrients would be circulated through subsurface soils and resulting elutriate would be collected for treatment at the groundwater treatment facility.	<u>Advantages</u> - o Process is performed in situ, eliminating the need for extensive soil excavation. <u>Disadvantages</u> - o Because contaminants extend to relatively deep levels (25 feet), implementation would require a subsurface circulation system (similar to soil flushing) to introduce and maintain microbes, nutrients, and oxygen supply. - o Typically would be coupled with biological treatment of groundwater.	<u>Advantages</u> <u>Disadvantages</u> - o The requirements for biological treatment make alternative will comparably more expensive than soil flushing. Both are theoretically equivalent in terms of effectiveness. - o Actual costs can not be determined prior to site-specific treatability tests.

Conclusion. Biological treatment of deep subsurface soils is most effective when coupled with the biological treatment of groundwater. Because biological treatment of groundwater has not been retained for detailed analysis, this alternative is also eliminated from detailed analysis.

5.3.2.5 Alternative SC-6: Low-temperature Thermal Stripping. This alternative consists of treating excavated soils contaminated with chemicals above TCLs, using a low-temperature thermal reactor to strip or desorb volatile organic chemicals from the soil. Treated soils would be backfilled on-site.

Effectiveness	Implementation	Cost
<u>Advantages</u>	<u>Advantages</u>	<u>Advantages</u>
- o Process can be controlled to maximize removal efficiencies. - o Would result in a clean soil which can be re-deposited on site. - o Sandy soils present are suitable for this process. - o Would substantially reduce the toxicity and volume of organic chemicals in soil. - o Highly effective on the chemicals of concern at both the Kysor and Four Winns sites. - o Remedy is permanent.	- o Openly accessible area is readily available at both the Kysor and Four Winns sites for set-up of process equipment. - o Equipment easily transportable to move between the Kysor and Four Winns sites.	o Lower capital costs than other forms of thermal treatment (on-site or off-site incineration).
<u>Disadvantages</u>	<u>Disadvantages</u>	<u>Disadvantages</u>
o Process produces air emissions that would require vapor phase carbon adsorption. Spent carbon would either be regenerated off-site or disposed of at a secured landfill.	- o Process air emissions would require adequate control. - o Requires the excavation affected soils. - o Damp or wet soils would require greater processing energies to desorb chemicals of concern.	o Additional costs would be associated with controlling air emissions.

Conclusion. Because the low-temperature thermal stripping alternative is technically suitable for both the Kysor and Four Winns sites, it would reduce the toxicity and volume of organic contaminants present in the soil, and it has the lowest capital cost of the thermal treatments, it is retained for detailed analysis.

5.3.2.6 Alternative SC-7: In-situ Soil Flushing. This alternative consists of in-situ soil flushing to remove the contaminants of concern from soil to concentrations below TCLs, collecting flushate using groundwater extraction wells, and treating the flushate using one of the groundwater treatment alternatives.

Effectiveness	Implementation	Cost
<u>Advantages</u> - o Effective for removal of chemicals from the zone of contamination by mimicking the natural infiltration. - o Most effective on water-soluble, water-mobile chemicals. - o Removes chemicals from soil thereby reducing the volume and mobility of contaminants in soil.	<u>Advantages</u> - o Process is performed in situ, eliminating the need for soil excavation. - o Sandy soils present at both Kysor and Four Winns is well suited to soil flushing.	<u>Advantages</u> - o This alternative would be cost effective if implemented in conjunction with a groundwater extraction/treatment alternative. - o Remedy is considered permanent.
<u>Disadvantages</u> - o May require the use of surfactants or additives to improve effectiveness. - o Potential for leaving contaminants in subsurface soils or groundwater if the collection system is not adequately designed. - o Flushate would need to be treated by one of the groundwater alternatives.	<u>Disadvantages</u> o Must be implemented in conjunction with a groundwater extraction/treatment alternative.	<u>Disadvantages</u> o None.

Conclusion. Soil flushing for treatment of soils at both the Kysor and Four Winns sites would be cost effective if implemented with a groundwater treatment alternative. Soil flushing is technically feasible and is retained for detailed analysis.

5.3.2.7 Alternative SC-8: Excavation/Solvent Extraction. This alternative consists of using a process reactor to mix excavated soils with a solvent (possibly water) to physically remove organic chemicals from excavated soils. Treated soils will be backfilled on-site.

Effectiveness	Implementation	Cost
<u>Advantages</u>	<u>Advantages</u>	<u>Advantages</u>
- o Would reduce the volume and mobility of contaminants in soil by removing chemicals from soil. - o Would result in a clean soil that can be re-deposited on-site. - o Remedy is considered permanent.	o Process equipment is readily available.	o None.
<u>Disadvantages</u>	<u>Disadvantages</u>	<u>Disadvantages</u>
- o May require the use of surfactants or additives to improve effectiveness. - o Potential for leaving contaminants or chemical by-products in soil if the solvent is not thoroughly removed. - o Process would produce solvents that would need to be treated either on-site at the groundwater treatment facility or off-site at a RCRA facility.	o Would require excavation of effected soils.	o While treatment element is conceptually similar to soil flushing, this alternative would result in higher costs due to excavation, dewatering, and material handling.

Conclusion. Because solvent extraction would require extensive excavation and has higher capital costs than the in-situ soil flushing alternative (SC-7) while achieving the same level of effectiveness, it is not retained for detailed analysis. If during the detailed evaluation of the soil flushing alternative, it becomes apparent soils must be excavated, the solvent extraction alternative will be reconsidered.

5.3.2.8 Alternative SC-9: In-situ Vacuum Extraction. This alternative consists of in-situ vacuum extraction to physically remove organic chemicals in soil from the unsaturated zone.

Effectiveness	Implementation	Cost
<u>Advantages</u> - o Most effective on VOCs with high vapor pressures (TCE, 111 TCA, and MC). May require longer treatment for xylene, which has a lower vapor pressure. - o Results in a reduction of volume of contaminants in soil.	<u>Advantages</u> - o Treatment process is performed in situ, eliminating soil excavation. - o Sandy soils at both the Kysor and Four Winns sites are well-suited to vacuum extraction.	<u>Advantages</u> - o In-situ process is cost effective. - o Remedy is considered permanent.
<u>Disadvantages</u> - o May require pilot-scale testing to determine site-specific effectiveness. - o Process produces air emissions that would require vapor phase carbon adsorption. Spent carbon would either be regenerated off-site or disposed of at a secured landfill.	<u>Disadvantages</u> - o May require capping to effectively remove near-surface contaminants. - o Process air emissions may require treatment, typically using activated carbon.	<u>Disadvantages</u> o None.

Conclusion: Because in-situ vacuum extraction would reduce the volume of contaminants in the soil at the Kysor and Winns sites, it is an in situ process, and it is cost effective, it is retained for detailed analysis.

5.4.2.9 Alternative SC-10: Off-site Incineration. This alternative consists of the excavation of soils with contaminants exceeding established TCLs, containerization, and transport to a commercial off-site RCRA-permitted incinerator for treatment and disposal.

Effectiveness	Implementation	Cost
<u>Advantages</u>	<u>Advantages</u>	<u>Advantages</u>
o Would reduce toxicity, volume and mobility of chemicals in soil through thermal destruction.	o Would eliminate the need to construct and operate process equipment at the site.	o Long-term monitoring would not be required.
o Pilot-scale tests have demonstrated removal efficiencies of 99.9999 percent for a variety of organic compounds.	o Currently, commercial RCRA-permitted incineration facilities are not available in State of Michigan. Commercial, RCRA-permitted incineration facilities exist near Chicago, Illinois, and St. Louis, Missouri.	
o Would remove the contaminated soil from the site.		
o Remedy is considered permanent.		
<u>Disadvantages</u>	<u>Disadvantages</u>	<u>Disadvantages</u>
o Transport of contaminated soil would increase the potential risk of exposure by the public.	o Currently, the availability of an off-site facility with the capacity to handle large quantities of soil is very limited.	o Relatively high cost for a treatment alternative.
	o Excavated soils must be transferred to small containers (i.e., 55-gallon drums) prior to acceptance by an off-site facility.	o Transport of soil to out-of-state facilities will substantially increase costs.

Conclusion. Because of the potential for a large volume of material to be treated at each site, the associated transportation requirements, limited facility capacity, facility acceptance of only containerized materials, and the high costs associated with off-site incineration, this alternative is not retained for detailed analysis.

5.3.2.10 Alternative SC-11: On-site Incineration. This alternative consists of excavation of contaminated soils for destruction of organic contaminants in an on-site incinerator, utilizing either a rotary-kiln, infrared, or fluidized bed incinerator. Ash passing the EP Toxicity test will be backfilled on-site.

Effectiveness	Implementation	Cost
<u>Advantages</u>	<u>Advantages</u>	<u>Advantages</u>
- o Would reduce toxicity, volume, and mobility of chemicals in soil through thermal destruction. - o Pilot-scale tests have demonstrated removal efficiencies of 99.9999 percent for a variety of organic compounds. - o Remedy is considered permanent.	- o Considered practical for waste volumes greater than 10,000 cy. - o Openly accessible area is readily available at both the Kysor and Four Winns sites for setup of an on-site incinerator.	o Long-term monitoring is not required.
<u>Disadvantages</u>	<u>Disadvantages</u>	<u>Disadvantages</u>
- o A trial burn is required to demonstrate compliance with emission limits. - o If ash does not pass the EP Toxicity test, it would have to be solidified. - o Scrubber effluent would be disposed of at off-site RCRA facility.	- o The incinerator would have to be setup at both Kysor and Four Winns. - o Ash must be tested for EP Toxicity prior to on-site disposal. - o Air emissions must be controlled.	- o Relatively high cost for a treatment alternative. - o If solidification of ash is required due to residual metals content, costs would increase significantly.

Conclusion. Because on-site incineration is technically feasible, would reduce toxicity, volume, and mobility of contaminants in the soil, and is practical for the volume of waste to be treated, it is retained for detailed analysis.

5.4 SUMMARY OF SCREENING OF REMEDIAL ALTERNATIVES FOR SOIL

The range of soil remedial alternatives retained after screening for detailed evaluation in the FS are presented as follows. Each alternative will be considered and evaluated for both the Kysor and Four Winns sites.

- o Alternative SC-1: No-action
- o Alternative SC-2: Surficial Capping/Subsurface
Containment Wall
- o Alternative SC-6: Low-temperature Thermal Stripping
- o Alternative SC-7: In-situ Soil Flushing
- o Alternative SC-8: In-situ Vacuum Extraction
- o Alternative SC-11: On-site Incineration

GLOSSARY OF ACRONYMS

ARARs	Applicable or Relevant and Appropriate Requirements
AWQC	Ambient Water Quality Criteria
BACT	best available control technology
BAT	best available technology
BDATs	best demonstrated available treatment technologies
BPJ	best professional judgement
CAG	Carcinogen Assessment Group
CDI	Chronic Daily Intake
CERCLA	Comprehensive Environmental Response, Compensation, and Liability Act of 1980 (the Superfund statute)
CRDLs	Contract Required Detection Levels
CWA	Clean Water Act
cy	cubic yards
1,2-DCA	1,2-dichloroethane
1,1-DCE	1,1-dichloroethylene
DNAPL	dense non-aqueous phase liquid
FS	Feasibility Study
HEA	Health Effects Assessment
HSWA	Hazardous and Solid Waste Amendments
MC	methylene chloride
MCL	Maximum Contaminant Level
MCLG	Maximum Contaminant Level Goal
MDNR	Michigan Department of Natural Resources
mg/l	milligrams per liter
MGWQ	Michigan Groundwater Quality Rules

NAAQs	National Ambient Air Quality Standards
NCP	National Contingency Plan
NESHAPs	National Emission Standards for Hazardous Air Pollutants
NPDES	National Pollutant Discharge Elimination System
NPL	National Priority List
OLM	Organic Leaching Model
OSHA	Occupational Safety and Health Administration
PCE	tetrachloroethylene
POTWs	Publicly Owned Treatment Works
ppm	parts per million
RA	Risk Assessment
RI	Remedial Investigation
ROD	Record of Decision
SARA	Superfund Amendments and Reauthorization Act of 1986
SDWA	Safe Drinking Water Act
t-1,2-DCE	trans-1,2-dichloroethylene
TBC	To Be Considered
1,1,1-TCA	1,1,1-trichloroethane
TCE	trichloroethylene
TCLs	target clean-up levels
TCLP	Toxicity Characteristic Leaching Procedure
USEPA	United States Environmental Protection Agency
UV	Ultraviolet
VOCs	Volatile Organic Compounds

APPENDIX A
CHEMICAL-SPECIFIC ARARS

APPENDIX A
CHEMICAL-SPECIFIC ARARS
CADILLAC FEASIBILITY STUDY

SITE FEATURE	REQUIREMENT	REQUIREMENT SYNOPSIS	CONSIDERATION IN THE FS
GROUNDWATER			
Federal Applicable Requirements	Safe Drinking Water Act (SDWA) - Maximum Contaminant Levels (MCLs) (40 CFR 141.11 - 141.16)	MCLs have been promulgated for a number of common organic and inorganic contaminants. These levels regulate the concentration of contaminants in public drinking water supplies, and are considered applicable for groundwater aquifers used for drinking water.	When the risks to human health due to consumption of groundwater are assessed, concentrations of contaminants should be compared to their MCLs.
Federal Relevant and Appropriate Requirements	SDWA - Maximum Contaminant Level Goals (MCLGs)	MCLGs are health-based criteria that are relevant and appropriate for drinking water sources, as a result of SARA, when extraordinary risks exist. These goals are available for a number of organic and inorganic contaminants.	Groundwater concentrations of noncarcinogenic contaminants should be compared to their MCLGs when extraordinary risks exist.
	Federal Ambient Water Quality Criteria (AWQC)	Federal AWQC are health-based criteria that have been developed for 95 carcinogenic and noncarcinogenic compounds.	AWQC should be used in characterizing human health risks due to contaminant concentrations in groundwater.
State Applicable Requirements	Michigan (MWRL), Act 245, R 323 Part 22, Groundwater Discharge Rules	Draft proposed rules to establish allowable discharge levels to protect Michigan groundwater quality in all usable aquifers as potential water supplies.	The draft revisions to the existing Part 22 Rules are still in the approval process; it is unknown when they will take effect.
Federal Advisories and Guidance to be Considered	Health Advisories (USEPA Office of Drinking Water)	Health advisories are estimates of risks due to consumption of contaminated drinking water; they consider noncarcinogenic effects only.	Health advisories should be considered for contaminants in groundwater that is used for drinking water.
	USEPA Risk Reference Doses (RfDs)	RfDs are dose levels developed by USEPA for noncarcinogenic effects.	USEPA RfDs are used to characterize risks due to exposure to contaminants in groundwater, as well as other media.
	USEPA Carcinogen Assessment Group Potency Factors	Potency Factors are developed by USEPA from Health Effects Assessments (HEA) or evaluations by the Carcinogen Assessment Group (CAG).	USEPA Carcinogenic Potency Factors are used to complete the individual incremental cancer risk resulting from exposure to contaminants.

APPENDIX A (continued)
CHEMICAL-SPECIFIC ARARS

CADILLAC FEASIBILITY STUDY

SITE FEATURE	REQUIREMENT	REQUIREMENT SYNOPSIS	CONSIDERATION IN THE FS
	Acceptable intake - Chronic (AIC) and Subchronic (AIS) - USEPA Health Assessment Documents	AIC and AIS values are developed from RfDs and HEAs for noncarcinogenic compounds.	AIC and AIS values are used to characterize the risks due to several noncarcinogens in various media.
	USEPA Office of Water Guidance - Water Related Fate of 129 Priority Pollutants (1979)	This guidance manual gives transport and fate information for 129 priority pollutants.	The manual is used to assess the transport and fate of a variety of contaminants.
<u>DISCHARGE TO PUBLICLY OWNED TREATMENT WORKS</u>			
Federal Applicable Requirements	CWA - Pretreatment Standards (40 CFR 403)	Discharges to a POTW must comply with the POTW's USEPA-approved pretreatment requirements.	POTWs in the area with approved pretreatment programs will be identified. The discharge must be treated to those levels required by the program.
State Applicable Requirements	MDRL, Act 245, R 323.2163 Pretreatment Program	Discharges to a POTW must comply with the POTW's pretreatment requirements.	POTWs in the area with approved pretreatment programs will be identified. The discharge must be treated to those levels required by the program.
<u>DISCHARGE TO SURFACE WATER/CLAM RIVER</u>			
State Applicable Requirements	Michigan Water Quality Standards, Act 245, R 323.1041-323.1100, R323.2137.	Standards have been developed for dissolved oxygen and temperature. Allowable levels of toxic substances will be determined by MDNR using appropriate scientific data. Water quality-based effluent limits will be used only if they are more restrictive than technology-based limits.	Technology-based standards will be used for setting limits for discharge to surface water.
Federal Relevant and Appropriate Requirements	Federal Ambient Water Quality Criteria (AWQC)	Federal AWQC are health-based criteria that have been developed for 95 carcinogenic and noncarcinogenic compounds.	AWQC may be used for setting limits for discharge to surface water.